



Chapter 18: Temperature

Temperature and 0th Law of Thermodynamics

Thermometers and Celsius Scale

Constant volume gas thermometer

Thermal expansion of solids and liquids

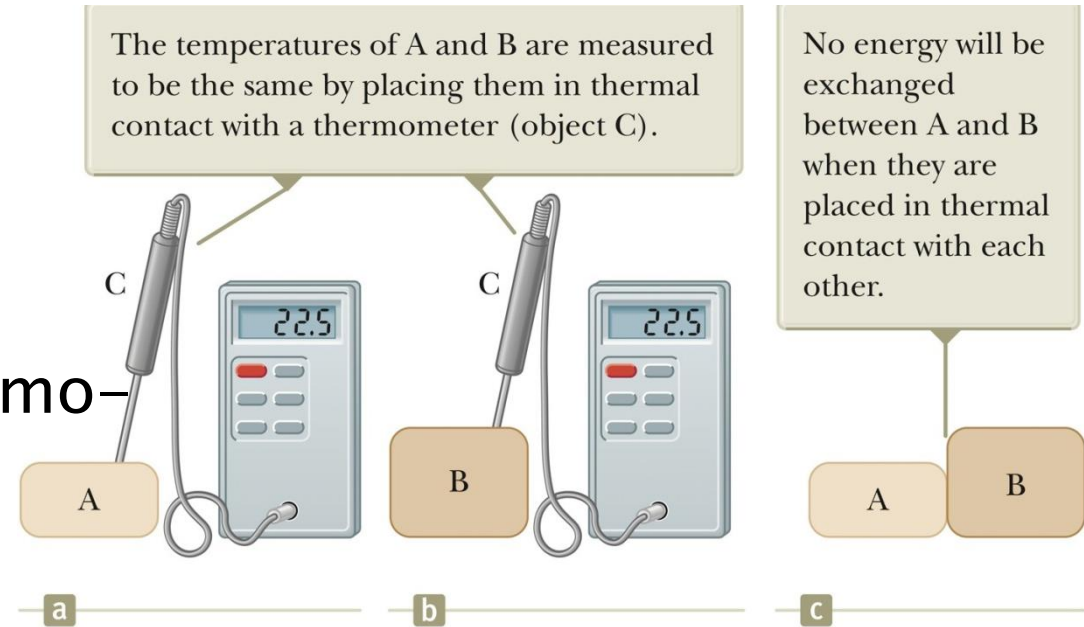
Macroscopic description of an ideal gas

Temperature

- ▶ Why does a tile floor feel colder than a carpet on a tile floor – even though they are at the same temperature?
- ▶ Our skin measures the rate of transfer of heat – Tile transfers heat faster than the carpet.
- ▶ So, we need a more reliable, reproducible method to measure temperature.
- ▶ **Thermal equilibrium** – two objects (can be at different temperatures) in thermal contact with each other will eventually reach the same final temperature due to energy exchanged between the two objects.

Temperature and 0th Law of thermodynamics

- ▶ Consider two objects *A* and *B*.
- ▶ Not in thermal contact.
- ▶ Third object *C*, the thermometer, is in contact with each object until in thermal equilibrium.
- ▶ If the thermometer indicated the same temperature for both, then objects *A* and *B* are in thermal equilibrium.
- ▶ **Zeroth Law:** If objects *A* and *B* are separately in thermal equilibrium with a third object *C*, then objects *A* and *B* are in thermal equilibrium with each other.



Quick Quiz 18.1

Two objects, with different sizes, masses, and temperatures, are placed in thermal contact. In which direction does the energy travel?

- (a) Energy travels from the larger object to the smaller object.
- (b) Energy travels from the object with more mass to the one with less mass.
- (c) Energy travels from the object at higher temperature to the object at lower temperature.

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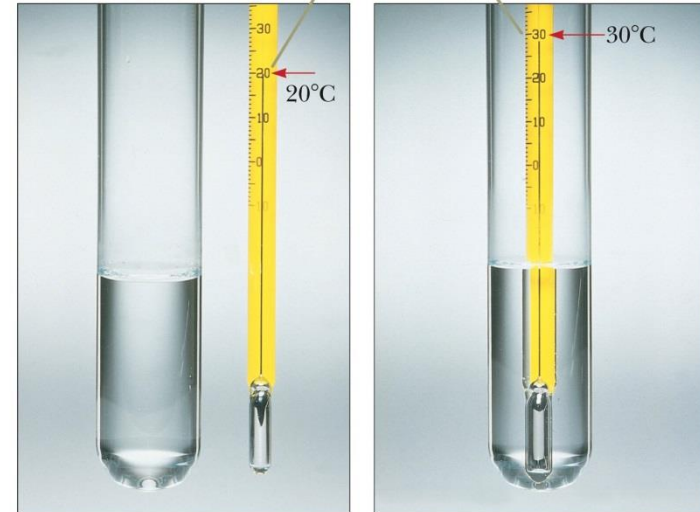
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Temperature and Celsius Scale

Some physical properties that change with temperature:

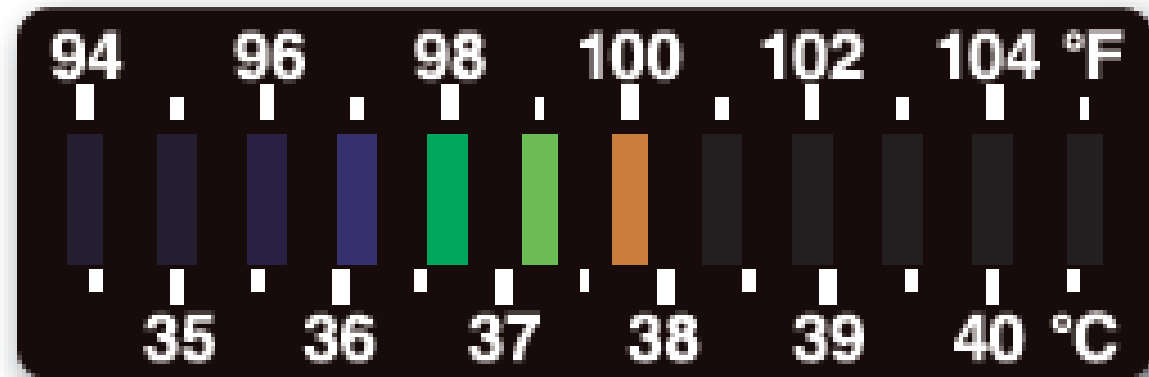
- Volume of liquid
- Dimensions of a solid
- Pressure of a gas at constant volume
- Volume of a gas at constant pressure
- Electrical resistance of a conductor
- Colour of and object.

The level of the mercury in the thermometer rises as the mercury is heated by water in the test tube.



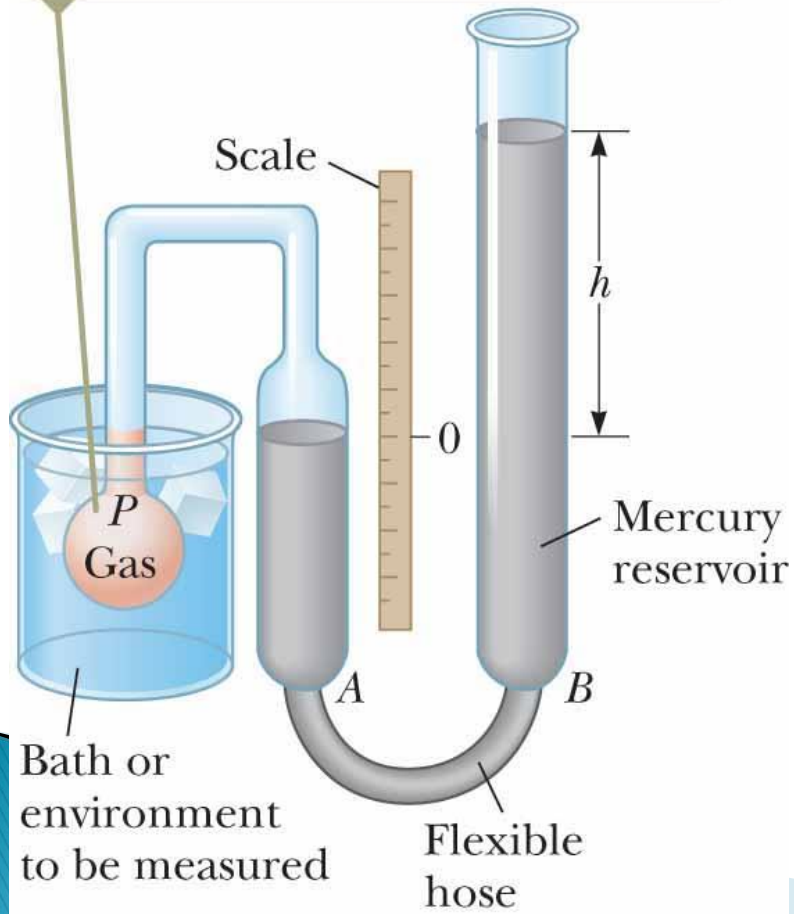
Common thermometers

- Calibrated on Celsius scale @ 2 pts.
- 0°C – water and ice in thermal equilibrium.
- 100°C – water and steam in thermal equilibrium.



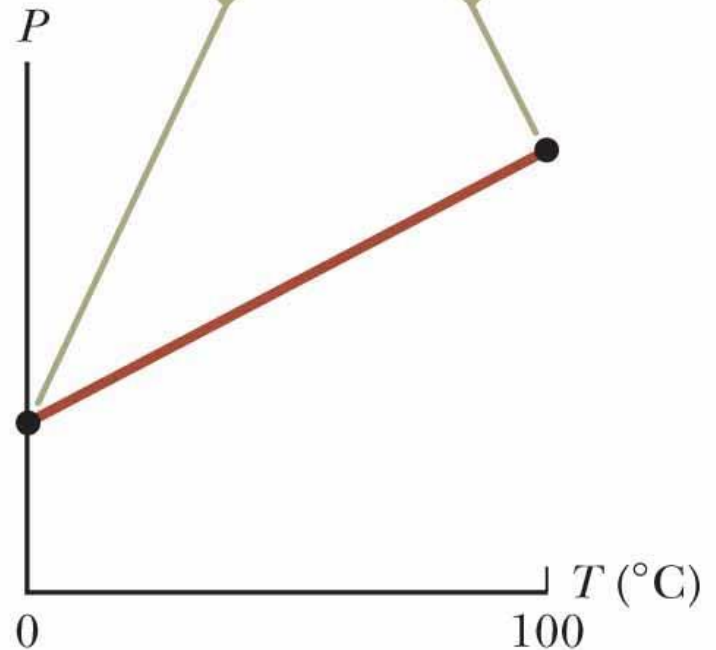
Constant-Volume Gas Thermometer

The volume of gas in the flask is kept constant by raising or lowering reservoir *B* to keep the mercury level in column *A* constant.



$$P = P_0 + \rho gh$$

The two dots represent known reference temperatures (the ice and steam points of water).



Constant-Volume Gas Thermometer

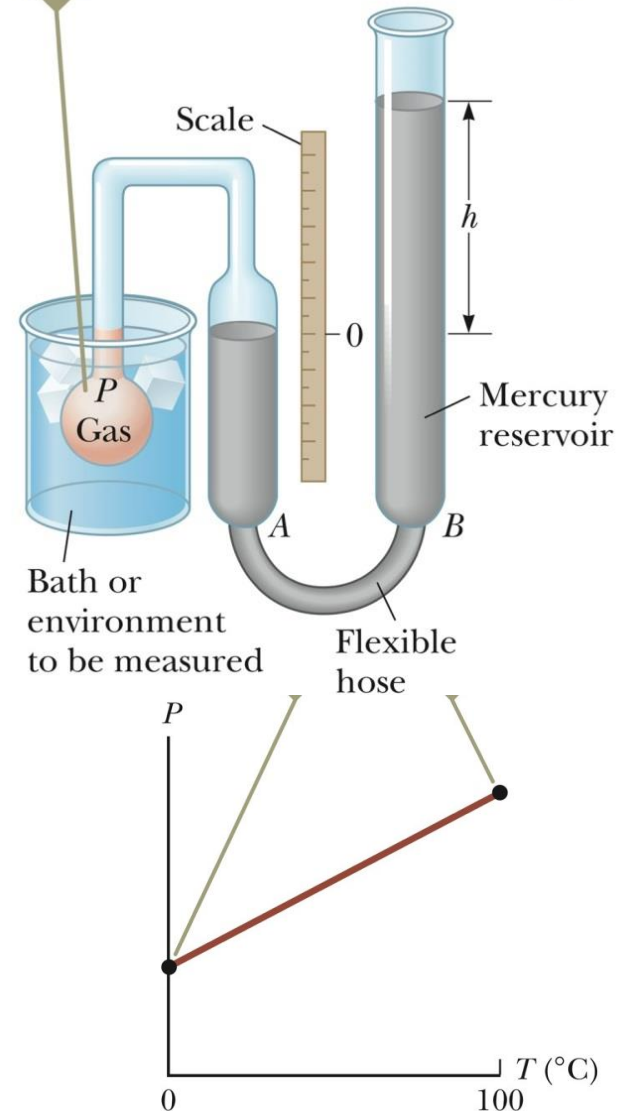
Read through!

- ▶ Flask immersed in ice–water
- ▶ Adjust column A to the zero height.
- ▶ Measure the height h difference between column A and B
- ▶ Indication of the pressure in flask.

$$P = P_0 + \rho gh$$

- ▶ Immerse flask in boiling water.
- ▶ Adjust column A to the zero height, and measure the height h difference between column A and B
- ▶ Draw straight line graph between 2 points and read off any temperature.

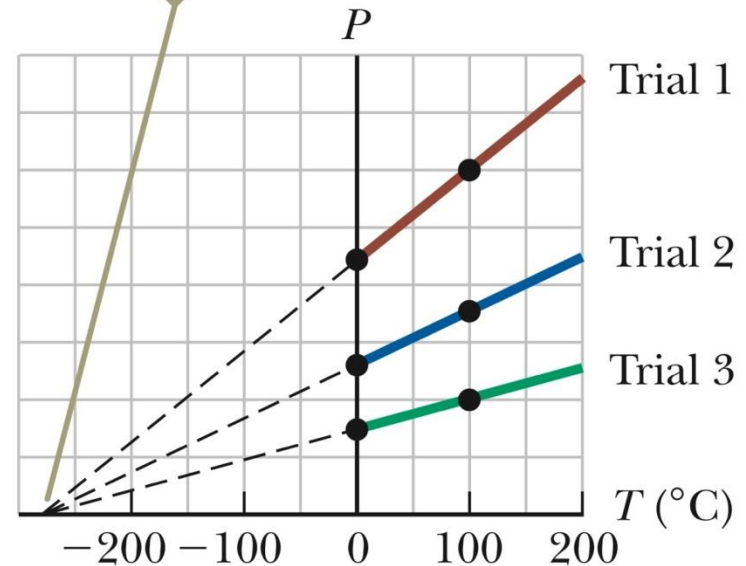
The volume of gas in the flask is kept constant by raising or lowering reservoir B to keep the mercury level in column A constant.



Constant-Volume Gas Thermometer

- ▶ Different gases
 - Between 0°C and 100°C
 - Extrapolate below 0°C
 - Pressure is zero when temperature is -273.15°C .
 - **Absolute zero** on the absolute temperature scale.
- ▶ SI unit: Kelvin – K
- ▶ Other units:

For all three trials, the pressure extrapolates to zero at the temperature -273.15°C .



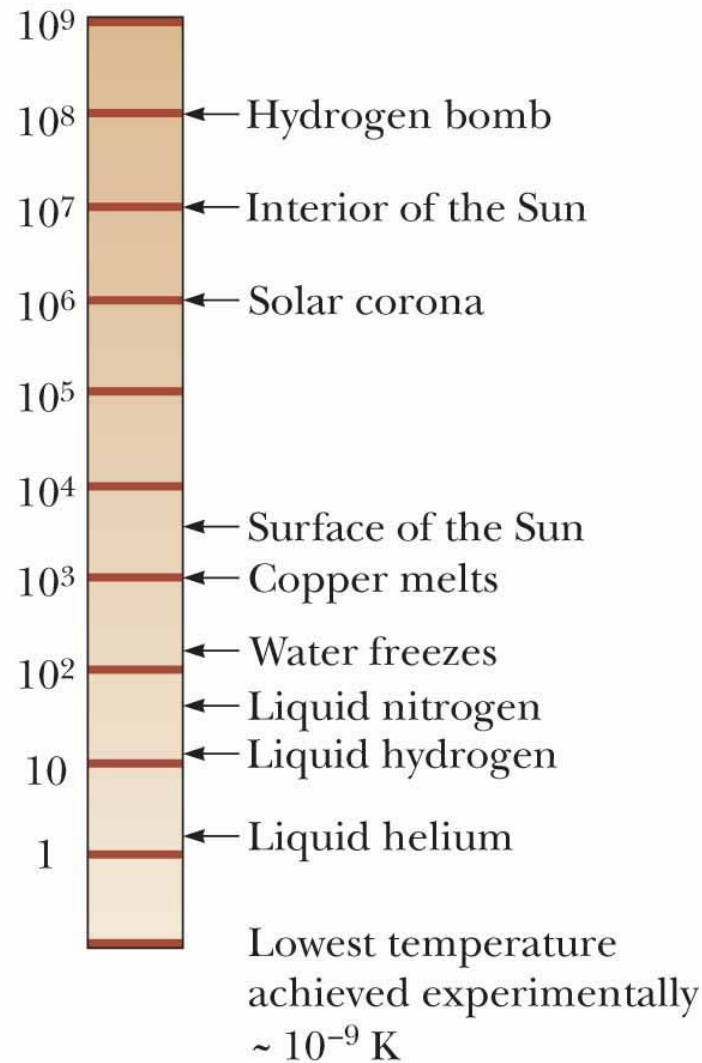
$$T_C = (T_K - 273.15)$$

$$T_F = \frac{9}{5} T_C + 32^{\circ}$$

The Absolute Temperature Scale

Note that the scale is logarithmic.

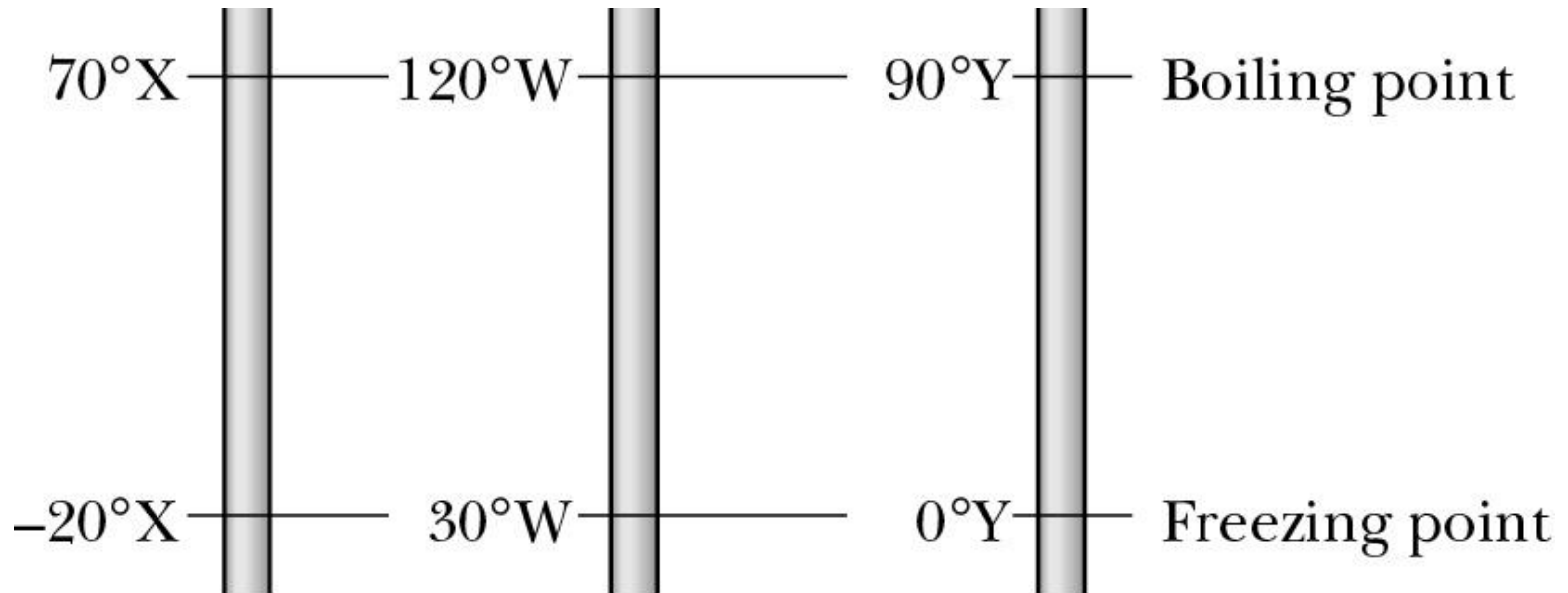
Temperature (K)



Problem

- ▶ The figure shows 3 linear temperature scales with freezing and boiling points of water indicated.

Q: Rank the degrees on these scales by size.



ANS: All are the same size

Quick Quiz 18.2

Consider the following pairs of materials. Which pair represents two materials, one of which is twice as hot as the other?

- (a) boiling water at 100°C , a glass of water at 50°C
- (b) boiling water at 100°C , frozen methane at -50°C
- (c) an ice cube at -20°C , flames from a circus fire-eater at 233°C
- (d) none of those pairs

Quick Quiz 18.2

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- (c) an ice cube at -20°C , flames from a circus fire-eater at 233°C**
- (d) none of those pairs

The phrase “twice as hot” refers to a ratio of temperatures. When the given temperatures are converted to kelvin, only those in part (c) are in the correct ratio.

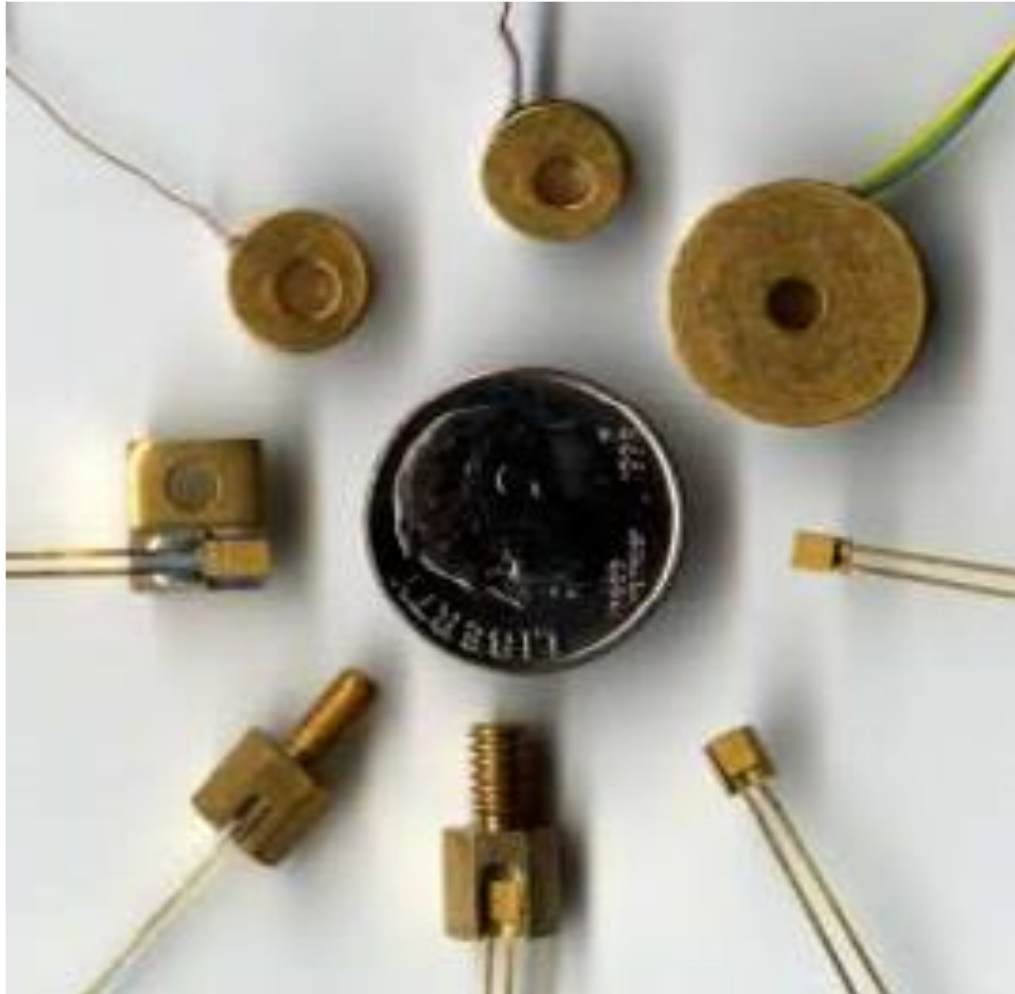


Temperature, Heat and 1st Law of Thermodynamics

Temperature 0th Law of Thermodynamics
Thermal expansion Heat and Work
1st Law of Thermodynamics

Thermal properties

- ▶ Thermocouples

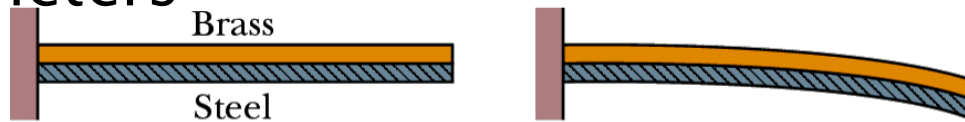


Thermal Expansion

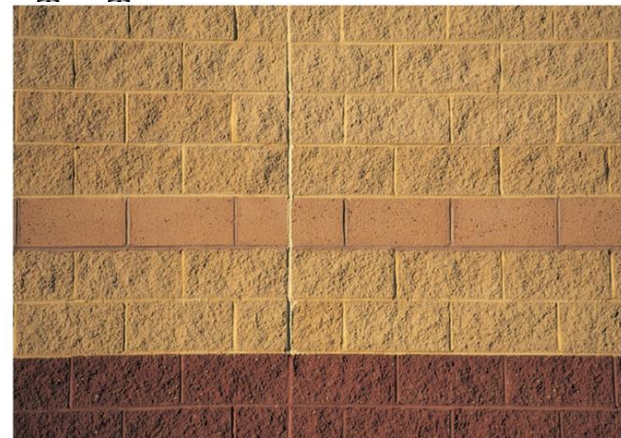
- ▶ What happens when the temperature of a material increases?
 - Energy is added
 - Atoms move a bit further apart
 - Different materials' atoms move different distances further apart

Thermal Expansion

- ▶ Thermal expansion must be taken into account in everyday situations or used to our advantage.
 - Bridge, train tracks and buildings have expansion joints
 - Dental fillings
 - Concorde (due to frictional forces)
 - Bimetal strip
 - Thermostats
 - Thermometers



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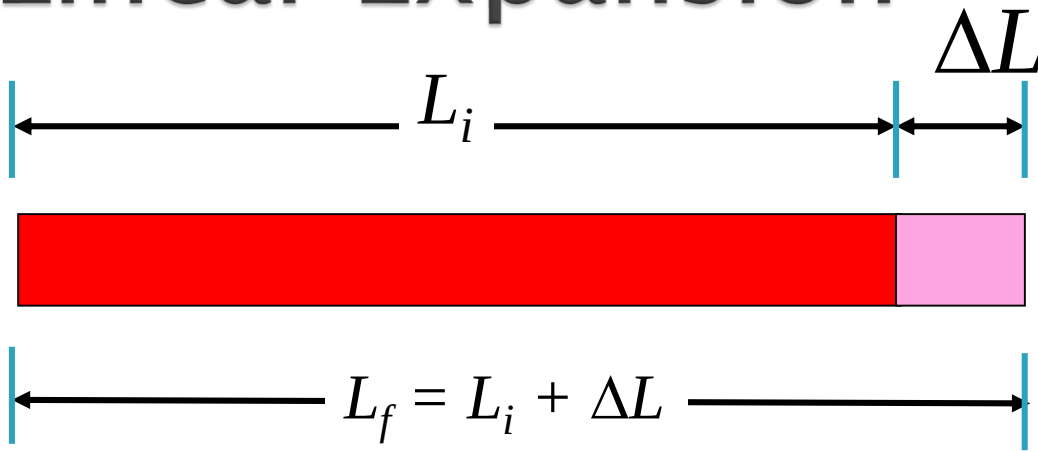


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Thermal Expansion

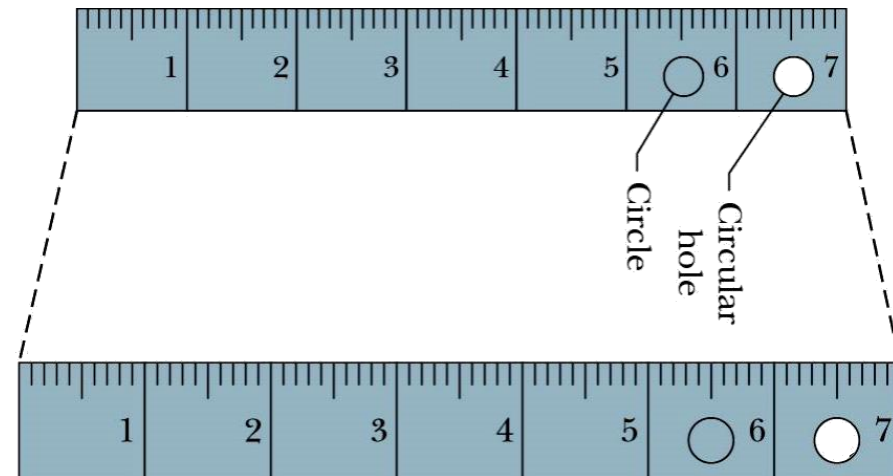


Linear Expansion



$$\Delta L = L_i \alpha \Delta T$$

- ▶ Consider a metal rod of length L whose temperature is raised by ΔT .
- ▶ Its **length will increase** from L_i to L_f :
- ▶ α is the **coefficient of linear expansion**
- ▶ Units of α : $^{\circ}\text{C}^{-1}$ or K^{-1}
- ▶ How do holes and voids in objects expand?
 - Circle
 - Scale
 - Numbers
 - Holes



Volume Expansion

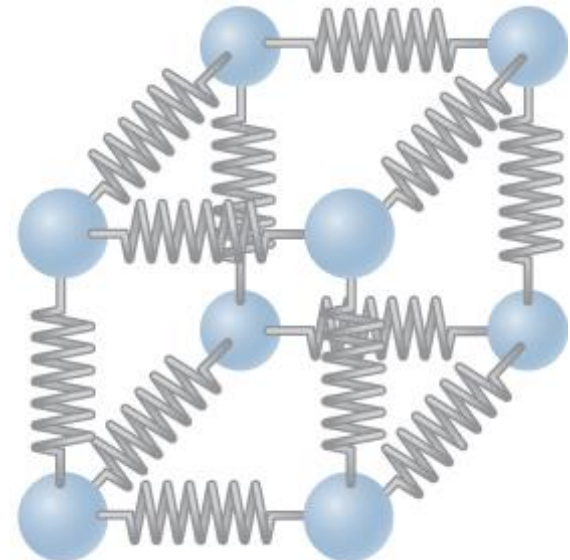
- ▶ All dimensions increase with an increase in temperature.
- ▶ Volume must also increase.

$$\Delta V = V \beta \Delta T$$

- ▶ β is the **coefficient of volume expansion**

$$\beta = 3\alpha$$

- ▶ Table 18.1



Thermal expansion

TABLE 18.1 Average Expansion Coefficients for Some Materials Near Room Temperature

Material (Solids)	Average Linear Expansion Coefficient (α)($^{\circ}\text{C}$) $^{-1}$	Material (Liquids and Gases)	Average Volume Expansion Coefficient (β)($^{\circ}\text{C}$) $^{-1}$
Aluminum	24×10^{-6}	Acetone	1.5×10^{-4}
Brass and bronze	19×10^{-6}	Alcohol, ethyl	1.12×10^{-4}
Concrete	12×10^{-6}	Benzene	1.24×10^{-4}
Copper	17×10^{-6}	Gasoline	9.6×10^{-4}
Glass (ordinary)	9×10^{-6}	Glycerin	4.85×10^{-4}
Glass (Pyrex)	3.2×10^{-6}	Mercury	1.82×10^{-4}
Invar (Ni–Fe alloy)	0.9×10^{-6}	Turpentine	9.0×10^{-4}
Lead	29×10^{-6}	Air ^a at 0 $^{\circ}\text{C}$	3.67×10^{-3}
Steel	11×10^{-6}	Helium ^a	3.665×10^{-3}

^aGases do not have a specific value for the volume expansion coefficient because the amount of expansion depends on the type of process through which the gas is taken. The values given here assume the gas undergoes an expansion at constant pressure.

Quick Quiz 18.4

Two spheres are made of the same metal and have the same radius, but one is hollow and the other is solid. The spheres are taken through the same temperature increase. Which sphere expands more?

- (a) The solid sphere expands more.
- (b) The hollow sphere expands more.
- (c) They expand by the same amount.
- (d) There is not enough information to say.

Quick Quiz 18.4

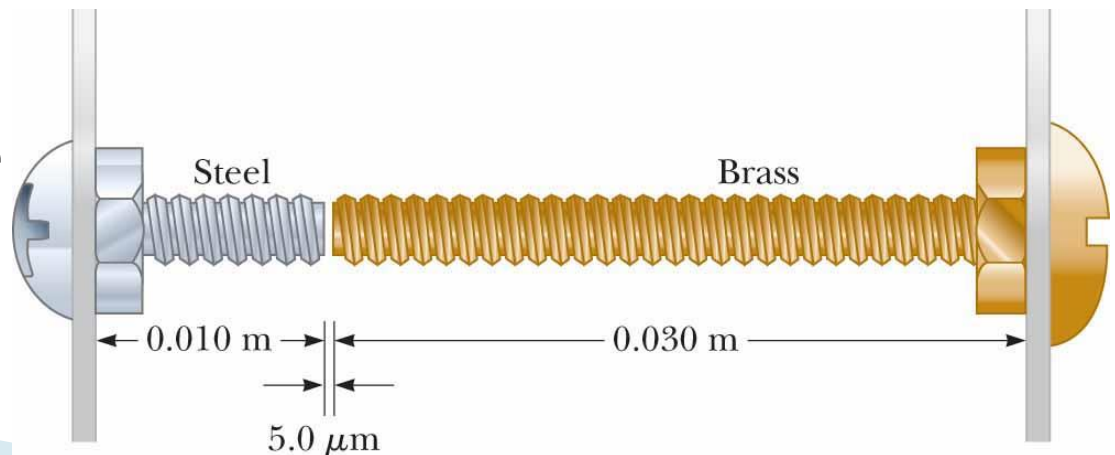
Two spheres are made of the same metal and have the same radius, but one is hollow and the other is solid. The spheres are taken through the same temperature increase. Which sphere expands more?

- (a) The solid sphere expands more.
- (b) The hollow sphere expands more.
- (c) They expand by the same amount.**
- (d) There is not enough information to say.

A cavity in a material expands in the same way as if it were filled with that material.

Example 18.3: The Thermal Electrical Short

A poorly designed electronic device has two bolts attached to different parts of the device that almost touch each other in its interior as in the figure. The steel and brass bolts are at different electric potentials, and if they touch, a short circuit will develop, damaging the device. (We will study electric potential in Chapter 25.) The initial gap between the ends of the bolts is $d = 5.0 \text{ mm}$ at 27°C . At what temperature will the bolts touch? Assume the distance between the walls of the device is not affected by the temperature change.



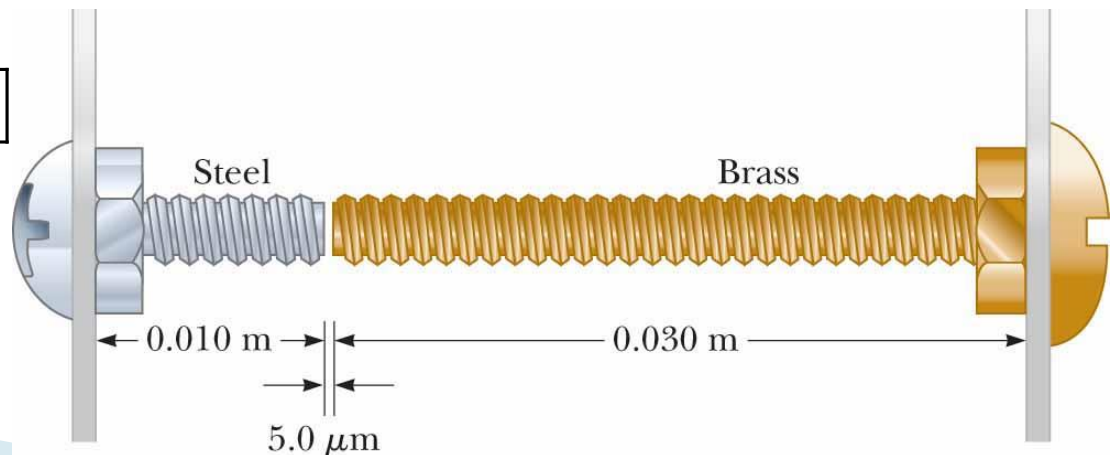
Example 18.3: The Thermal Electrical Short

$$\Delta L_{\text{br}} + \Delta L_{\text{st}} = \alpha_{\text{br}} L_{i,\text{br}} \Delta T + \alpha_{\text{st}} L_{i,\text{st}} \Delta T = d$$

$$\Delta T = \frac{d}{\alpha_{\text{br}} L_{i,\text{br}} + \alpha_{\text{st}} L_{i,\text{st}}}$$

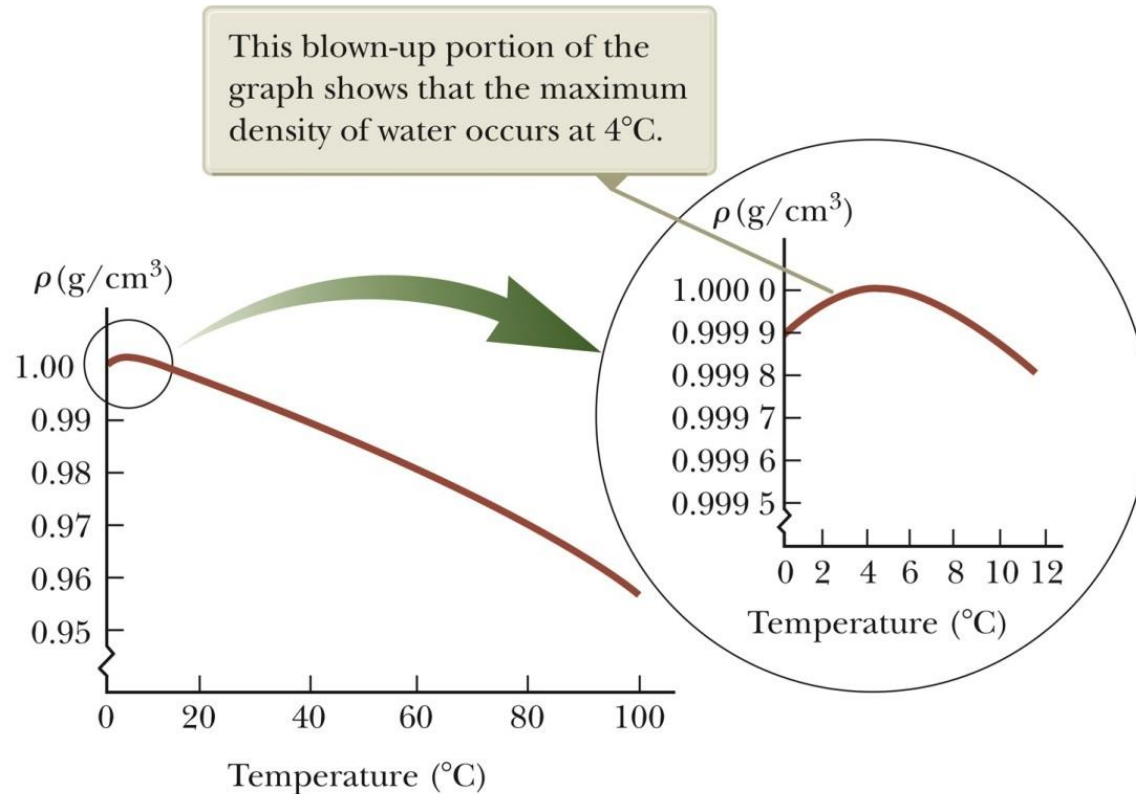
$$\Delta T = \frac{5.0 \times 10^{-6} \text{ m}}{\left[19 \times 10^{-6} (\text{°C})^{-1}\right](0.030 \text{ m}) + \left[11 \times 10^{-6} (\text{°C})^{-1}\right](0.010 \text{ m})} = 7.4\text{°C}$$

$$\Delta T = 27\text{°C} + 7.4\text{°C} = \boxed{34\text{°C}}$$



Why does ice float in water?

- ▶ Above 4 °C water expands with an increase in temperature like other solids – density decreases.
- ▶ Between 0 to 4 °C water contracts with increase in temperature – density increases.
- ▶ At 4 °C water has a max. density.

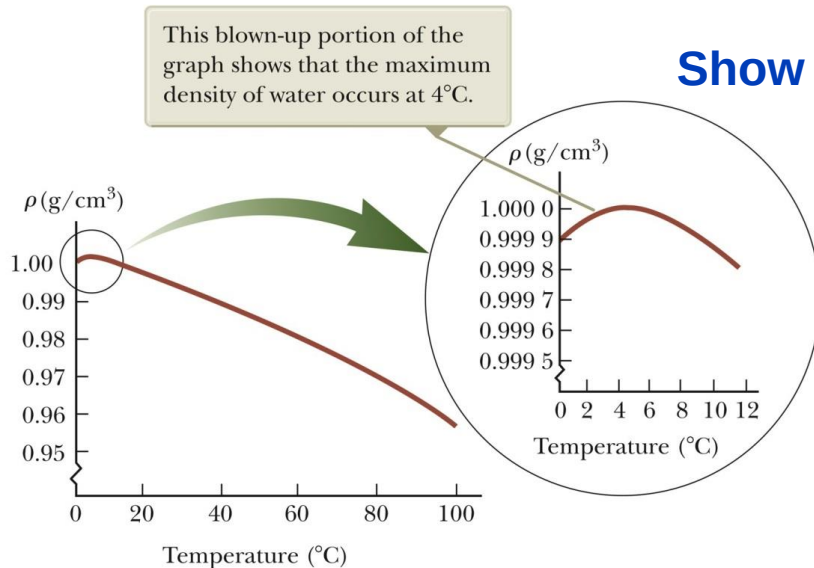


Why does ice float in water?



- Pond begins freezing at surface rather than at bottom
- When air temperature drops from, for example, 7°C to 6°C :
 - Surface water cools $\rightarrow$ decreases in volume
- The surface water denser than water below it
 - Surface water sinks
 - Warmer water from below moves to surface
- Air temperature between 4°C and 0°C :
 - Surface water expands as it cools $\rightarrow$ becoming less dense than water below it
 - Mixing process stops
 - Eventually surface water freezes
- As water freezes, ice remains on surface because ice is less dense than water
 - Ice continues to build up at surface
 - Water near the bottom remains at 4°C
- If that were not the case, fish and other forms of marine life would not survive

[Show video](#)



Problem

- A steel train track has a length of 30 m when the temperature is 32 °F. What is its length when the temperature is (a) 313 K? (b) -40°C?

(a) $\Delta T = 313 - 273 = 40 \text{ K}$

$$\Delta L = L\alpha\Delta T$$

$$= (30 \text{ m})(11 \times 10^{-6} / \text{C}^\circ)(40 \text{ C}^\circ)$$

$$= 0.0132 \text{ m}$$

$$L_f = L_i + \Delta L$$

$$= 30.000 + 0.0132 \text{ m}$$

$$= 30.0132 \text{ m}$$

(b) $\Delta T = -40 - 0 \text{ }^\circ\text{C} = -40 \text{ }^\circ\text{C}$

$$\Delta L = L\alpha\Delta T$$

$$= (30 \text{ m})(11 \times 10^{-6} / \text{C}^\circ)(-40 \text{ C}^\circ)$$

$$= -0.0132 \text{ m}$$

$$L_f = L_i + \Delta L$$

$$= 30.000 - 0.0132 \text{ m}$$

$$= 29.9868 \text{ m}$$

Problem

- On a hot day in LA, an oil trucker loaded 37 000 L of diesel fuel. He encountered cold weather on his journey, and where it was 23.0 K lower than in LA, he delivers his entire load. How many litres did he deliver? The coefficient of volume expansion for diesel is $9.50 \times 10^{-4}/\text{C}^\circ$, and the coefficient of linear expansion for the steel tank is $11 \times 10^{-6}/\text{C}^\circ$.

$$\begin{aligned}\Delta V &= V \beta \Delta T \\ &= (37000 \text{ L}) (9.5 \times 10^{-4} / \text{C}^\circ) (-23.0 \text{ K}) \\ &= -808 \text{ L}\end{aligned}$$

$$\begin{aligned}V_f &= V_i + \Delta V \\ &= (37000 \text{ L}) - 808 \text{ L} \\ &= 36192 \text{ L}\end{aligned}$$

Macroscopic description of an ideal gas

- ▶ Read section 18.5

Boltzmann's Constant

$$PV = nRT$$

$$PV = nRT = \frac{N}{N_A} RT$$

$$PV = Nk_B T$$

$$k_B = \frac{R}{N_A} = 1.38 \times 10^{-23} \text{ J/K}$$

- Ideal gas law often expressed in terms of total number of molecules N
- Because number of moles n = ratio of total number of molecules and Avogadro's number N_A ,
 - k_B = **Boltzmann's constant**:
- Common to call quantities such as P , V , and T **thermodynamic variables** of ideal gas
 - If equation of state known:
 - One of the variables can always be expressed as some function of other two



Chapter 19:1st Law of Thermodynamics

Heat and internal energy

Specific heat

Latent heat

Work and heat in thermodynamic processes

1st Law of thermodynamics

Energy transfer mechanisms

Heat and internal energy

- ▶ **Internal energy** is all the energy of a system that is associated with its microscopic components – atoms and molecules – when viewed from a reference frame at rest with respect to the COM of the system.
- ▶ **Heat Q** : A Process of transferring energy across the boundary of a system due to a temperature difference between the system and its surroundings.
- ▶ Heat is also the amount of energy, Q , transferred by this process – measured in Joule
- ▶ Other common units:
 - 1 calorie (cal) = 4.1860 J
 - 1 Calorie (Cal) = 1 kcal = 4 186.0 J [Diet]

J.P. Joule 1818 – 1889

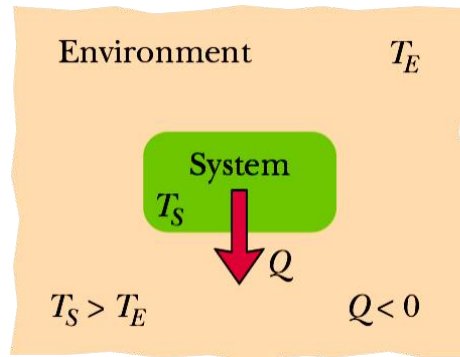


calorie (cal): energy transfer necessary to raise temperature of 1 g of water from 14.5°C to 15.5°C

Heat

▶ $T_S > T_E$

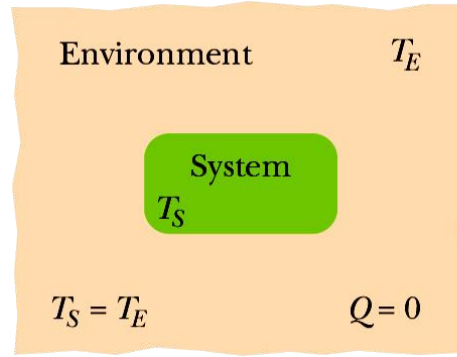
▶ $Q < 0$



(a)

$$T_S = T_E$$

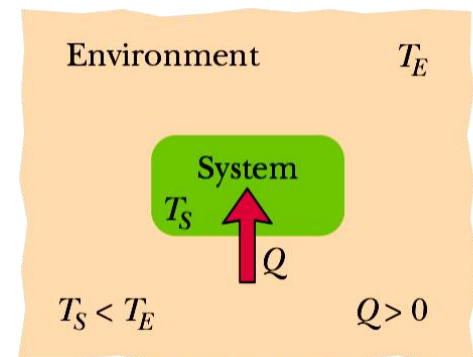
$$Q = 0$$



(b)

$$T_S < T_E$$

$$Q > 0$$



(c)

Sign convention:

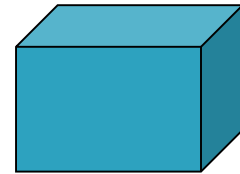
Flow out of system: $Q < 0$

Flow into system: $Q > 0$

Specific Heat and Calorimetry

- ▶ **Heat capacity C** – proportionality constant between Q and ΔT ; has units (J/K)

$$Q = C\Delta T = C(T_f - T_i)$$



- ▶ **Specific Heat (Capacity) c** – Heat capacity per unit mass (J/kg·K)

$$Q = cm\Delta T = cm(T_f - T_i)$$

Table 19.1

- ▶ **British thermal unit (Btu)** - Energy transfer required to raise temperature of 1 lb of water from 63°F to 64°F

Specific Heat and Calorimetry

TABLE 19.1 Specific Heats of Some Substances at 25°C and Atmospheric Pressure

Substance	Specific Heat (J/kg · °C)	Substance	Specific Heat (J/kg · °C)
<i>Elemental solids</i>		<i>Other solids</i>	
Aluminum	900	Brass	380
Beryllium	1 830	Glass	837
Cadmium	230	Ice (−5°C)	2 090
Copper	387	Marble	860
Germanium	322	Wood	1 700
Gold	129	<i>Liquids</i>	
Iron	448	Alcohol (ethyl)	2 400
Lead	128	Mercury	140
Silicon	703	Water (15°C)	4 186
Silver	234	<i>Gas</i>	
		Steam (100°C)	2 010

Note: To convert values to units of cal/g · °C, divide by 4 186.

Specific Heat and Calorimetry

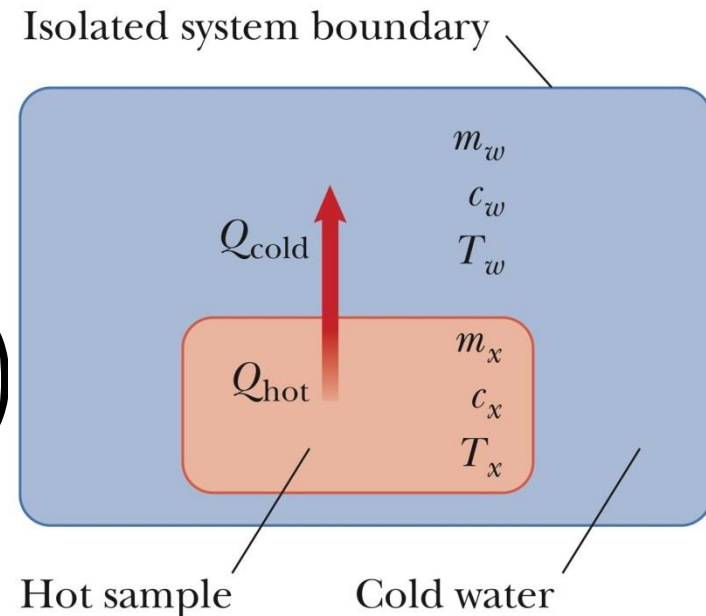
▶ Calorimetry –

- Heat an object to a specific temperature T_x
- Place the object in a container with a known amount of water at a known temperature $T_x > T_w$
- Measure the final equilibrium temperature of the water.

$$Q_{cold} = -Q_{hot}$$

▶ Solve for the unknown c_x

$$m_w c_w (T_f - T_w) = -m_x c_x (T_f - T_x)$$



Latent Heat

- ▶ **Latent Heat, L** – Heat absorbed by the sample when it changes phase – temperature is constant (J kg^{-1})

$$Q = Lm$$

- L_F heat of fusion (freezing or melting)
- L_V heat of vaporisation (boiling or condensing)

Table 19.2

Latent Heat

$$L \equiv \frac{Q}{\Delta m}$$

TABLE 19.2 Latent Heats of Fusion and Vaporization

Substance	Melting Point (°C)	Latent Heat of Fusion (J/kg)	Boiling Point (°C)	Latent Heat of Vaporization (J/kg)
Helium ^a	−272.2	5.23×10^3	−268.93	2.09×10^4
Oxygen	−218.79	1.38×10^4	−182.97	2.13×10^5
Nitrogen	−209.97	2.55×10^4	−195.81	2.01×10^5
Ethyl alcohol	−114	1.04×10^5	78	8.54×10^5
Water	0.00	3.33×10^5	100.00	2.26×10^6
Sulfur	119	3.81×10^4	444.60	3.26×10^5
Lead	327.3	2.45×10^4	1 750	8.70×10^5
Aluminum	660	3.97×10^5	2 450	1.14×10^7
Silver	960.80	8.82×10^4	2 193	2.33×10^6
Gold	1 063.00	6.44×10^4	2 660	1.58×10^6
Copper	1 083	1.34×10^5	1 187	5.06×10^6

^aHelium does not solidify at atmospheric pressure. The melting point given here corresponds to a pressure of 2.5 MPa.

Example

- ▶ A certain amount of heat Q will warm 1 g of material A by $3\text{ }^{\circ}\text{C}$ and 1 g of material B by $4\text{ }^{\circ}\text{C}$. Which material has the greater specific heat?

$$Q = cm\Delta T = cm(T_f - T_i)$$

$$c = \frac{Q}{m\Delta T}$$

A has the greater specific heat.

Example 19.3

A cowboy fires a silver bullet with a muzzle speed of 200 m/s into a wooden wall.

Assume all the internal energy generated by the impact remains with the bullet. What is the temperature change of the bullet?

Bullet is the system: isolated, with non-cons.

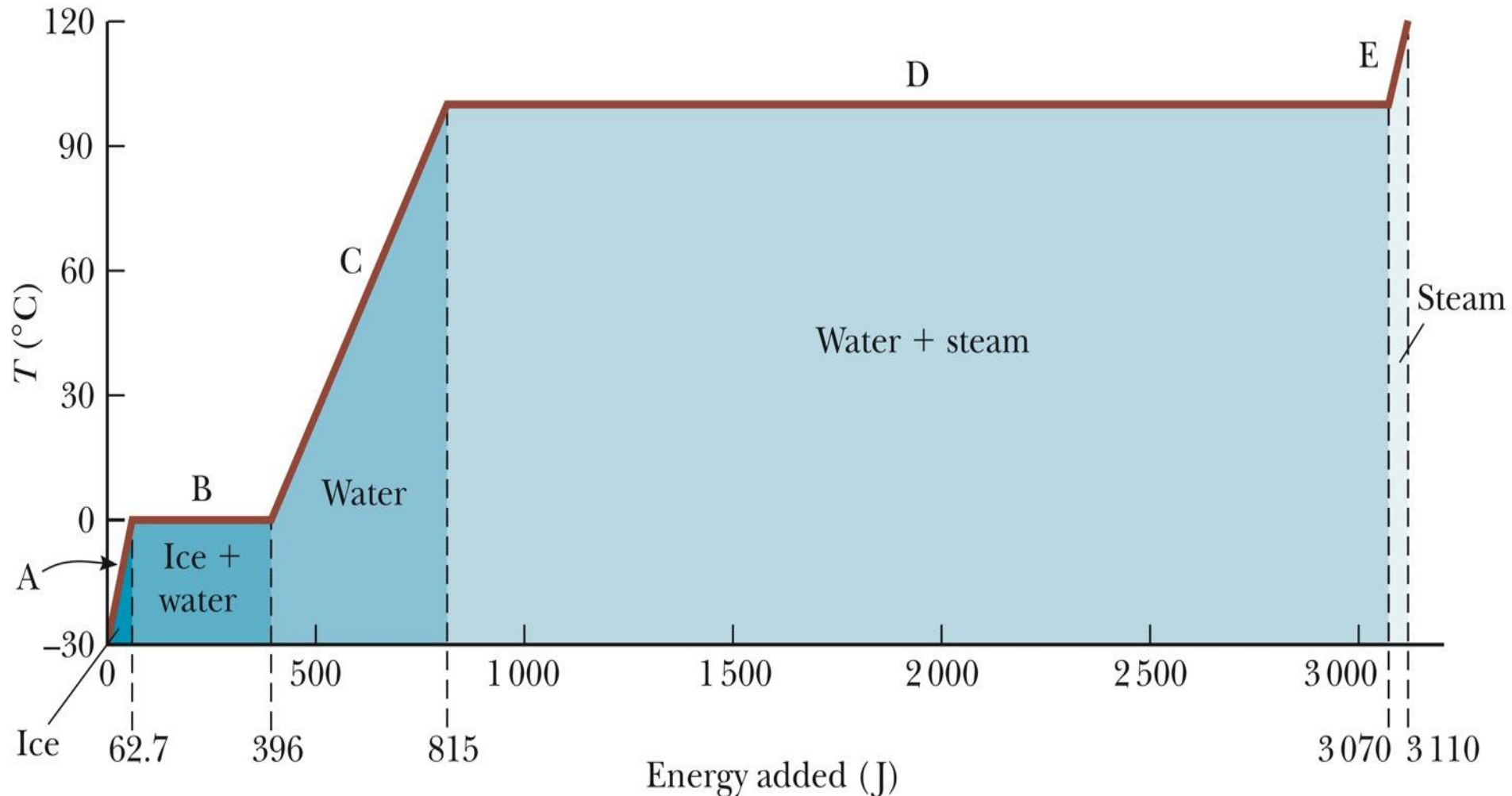
$$\Delta K + \Delta E_{\text{int}} = 0 \qquad -\frac{1}{2}mv^2 + mc\Delta T = 0$$

$$\Delta E_{\text{int}} = Q = cm\Delta T \qquad \Delta T = \frac{\frac{1}{2}mv^2}{mc} = \frac{v^2}{2c}$$

$$\Delta T = \frac{(200 \text{ m/s})^2}{2(234 \text{ J/kg}\cdot^\circ\text{C})} = 85.5^\circ\text{C}$$

Latent Heat

- ▶ 1 g of ice from -30°C to 120°C



Problem

- ▶ How much heat must be absorbed by ice of mass $m = 720 \text{ g}$ at -10°C to take it to liquid state at 15°C ?
- ▶ **3 step process:**
 - Raise ice temperature from -10°C to 0°C
 - Phase change ice to water at 0°C
 - Raise water temperature from 0°C to 15°C

$$Q = cm\Delta T = c_i m (T_f - T_i)$$

$$Q = L_F m$$

$$Q = cm\Delta T = c_w m (T_f - T_i)$$

Problem

- ▶ How much heat must be absorbed by ice of mass $m = 720 \text{ g}$ at -10°C to take it to liquid state at 15°C ?
- ▶ 3 step process:
 - Raise temperature from -10°C to 0°C $Q = cm\Delta T = c_i m(T_f - T_i)$
 - Phase change at 0°C – ice to water $Q = L_F m$
 - Raise temperature from 0°C to 15°C $Q = cm\Delta T = c_w m(T_f - T_i)$

$$Q_1 = c_i m \Delta T = c_i m (T_f - T_i)$$
$$= (2220 \text{ J/kg}\cdot\text{K})(0.720 \text{ kg})(0 - (-10)^\circ\text{C}) = 15.98 \text{ kJ}$$

$$Q_2 = L_F m = (333 \text{ kJ/kg})(0.720 \text{ kg}) = 239.8 \text{ kJ}$$

$$Q_3 = c_w m \Delta T = c_w m (T_f - T_i)$$
$$= (4190 \text{ J/kg}\cdot\text{K})(0.720 \text{ kg})(15 - 0^\circ\text{C}) = 45.25 \text{ kJ}$$

$$Q_T = Q_1 + Q_2 + Q_3 = 15.98 + 239.8 + 45.25 = 300 \text{ kJ}$$

Problem

If we supply the $-10\text{ }^{\circ}\text{C}$ ice with a total of only 210 kJ (as heat), what are the final temperature and state of the water?:

$$Q_1 = c_i m \Delta T = c_i m (T_f - T_i) \\ = (2220\text{ J/kg}\cdot\text{K})(0.720\text{ kg})(0 - (-10)\text{ }^{\circ}\text{C}) = 15.98\text{ kJ}$$

The ice is now at Temperature: $0\text{ }^{\circ}\text{C}$

Heat left (still available): $210\text{ kJ} - 15.98\text{ kJ} \approx 194\text{ kJ}$

How much water will melt with this amount of heat available?

$$Q_2 = L_F m = (333\text{ kJ/kg})(0.720\text{ kg}) = 239.8\text{ kJ}$$

$$Q_2 = L_F m$$

$$m = \frac{Q}{L_F} = \frac{194\text{ kJ}}{333\text{ kJ/kg}} = 0.580\text{ kg}$$

580 g is in
liquid state @ $0\text{ }^{\circ}\text{C}$

140 g is in
solid state (ice) @ $0\text{ }^{\circ}\text{C}$

Example 19.4: Cooling the Steam

What mass of steam initially at 130°C is needed to warm 200 g of water in a 100-g glass container from 20.0°C to 50.0°C ?

$$Q_{\text{cold}} = -Q_{\text{hot}}$$

$$Q_1 = m_s c_s \Delta T_s$$

$$\begin{aligned}\Delta m_s &= L_v (m_w - m_s) \\ &= L_v (0 - m_s) = -m_s L_v\end{aligned}$$

$$Q_2 = L_v \Delta m_s = L_v (0 - m_s) = -m_s L_v$$

$$Q_3 = m_s c_w \Delta T_{\text{hot water}}$$

$$Q_{\text{hot}} = Q_1 + Q_2 + Q_3 = m_s (c_s \Delta T_s - L_v + c_w \Delta T_{\text{hot water}})$$

$$Q_{\text{cold}} = m_w c_w \Delta T_{\text{cold water}} + m_g c_g \Delta T_{\text{glass}}$$

Example 19.4: Cooling the Steam

$$Q_{\text{cold}} = m_w c_w \Delta T_{\text{cold water}} + m_g c_g \Delta T_{\text{glass}}$$

$$m_w c_w \Delta T_{\text{cold water}} + m_g c_g \Delta T_{\text{glass}} = -m_s (c_s \Delta T_s - L_v + c_w \Delta T_{\text{hot water}})$$

$$m_s = - \frac{m_w c_w \Delta T_{\text{cold water}} + m_g c_g \Delta T_{\text{glass}}}{c_s \Delta T_s - L_v + c_w \Delta T_{\text{hot water}}}$$

$$\begin{aligned} m_s &= - \frac{(0.200 \text{ kg})(4186 \text{ J/kg} \cdot ^\circ\text{C})(50^\circ\text{C} - 20.0^\circ\text{C}) + (0.100 \text{ kg})(837 \text{ J/kg} \cdot ^\circ\text{C})(50.0^\circ\text{C} - 20.0^\circ\text{C})}{(210 \text{ J/kg} \cdot ^\circ\text{C})(100^\circ\text{C} - 30^\circ\text{C}) - (2.26 \times 10^6 \text{ J/kg}) + (4186 \text{ J/kg} \cdot ^\circ\text{C})(50.0^\circ\text{C} - 20.0^\circ\text{C})} \\ &= 1.09 \times 10^{-2} \text{ kg} = \boxed{10.9 \text{ g}} \end{aligned}$$

Example 19.4: Cooling the Steam

What if the final state of the system is water at 100°C? Would we need more steam or less steam? How would the analysis above change?

$$\Delta T_{\text{cold water}} = 100^{\circ}\text{C} - 20^{\circ}\text{C} = 80^{\circ}\text{C}$$

$$\Delta T_{\text{hot water}} = 100^{\circ}\text{C} - 100^{\circ}\text{C} = 0^{\circ}\text{C}$$

$$\Rightarrow m_s = 13.7 \text{ g}$$

Problem

- ▶ A copper slug whose mass m_c is 75 g is heated in a lab oven to 312 °C. The slug is then dropped into a glass beaker containing a mass $m_w = 220$ g of water. The heat capacity C_b of the beaker is 45 cal/K. The initial temperature T_i of the water and the beaker is 12 °C. Assuming the slug, beaker and the water are an isolated system and the water does not vaporize, find the final temperature T_f of the system at thermal equilibrium.

45 cal [4.1860J/cal]/K = 188.41 J/K CONVERSION TO J/K

	<u>Copper</u>	<u>water</u>	<u>beaker</u>
m (g)	75g	220 g	
C_b (J/K)			188.41
c (J/kg.K)	389	4180	
T_i (°C)	312	12	12
T_f (°C)	?	?	?

$$Q_c = c_c m_c (T_f - T_i) \quad Q_w = c_w m_w (T_f - T_i) \quad Q_b = C_b (T_f - T_i)$$

Problem

	<u>Copper</u>	<u>water</u>	<u>beaker</u>
m (g)	75g	220 g	
C_b (J/K)			188.41
C (J/kg.K)	389	4180	
T_i (°C)	312	12	12
T_f (°C)	?	?	?

$$Q_c = c_c m_c (T_f - T_i) \quad Q_w = c_w m_w (T_f - T_i) \quad Q_b = C_b (T_f - T_i)$$

$$\Delta Q = 0 \Rightarrow Q_{lost} \text{ (slug)} + Q_{gained} \text{ (water + beaker)}$$

$$-Q_{lost} = Q_{gained}$$

$$-Q_c = Q_w + Q_b$$

$$-c_c m_c (T_f - T_{i,c}) = c_w m_w (T_f - T_{i,w}) + C_b (T_f - T_{i,b})$$

$$T_f (-c_c m_c - c_w m_w - C_b) = (c_c m_c T_{i,c} - c_w m_w T_{i,w} - C_b T_{i,b})$$

$$T_f = \frac{-c_c m_c T_{i,c} - c_w m_w T_{i,w} - C_b T_{i,b}}{-c_c m_c - c_w m_w - C_b}$$

$$= 20 \text{ } ^\circ\text{C}$$

The End
tentative

Heat and work in a thermodynamic process

For any given configuration of the system, we can identify values of the **state variables**: P , V , T , E_{int} .

For mechanical systems, the **state variables** include kinetic energy K and potential energy U .

For a single particle as a system, we could identify more **state variables**: its position x , its velocity v , and its acceleration a .

$$\Delta K + \Delta U + \Delta E_{\text{int}} = W + Q + T_{MW} + T_{MT} + T_{ET} + T_{ER}$$

$$\Delta E_{\text{int}} = W + Q + \cancel{T_{MW} + T_{MT} + T_{ET} + T_{ER}}$$

Heat and work in a thermodynamic process

$$\Delta K + \Delta U + \Delta E_{\text{int}} = W + Q + T_{MW} + T_{MT} + T_{ET} + T_{ER}$$

Because a transfer of energy across the boundary represents a change in the system, transfer variables are not associated with a given state of the system, but rather with a change in the state of the system. (see PV diagrams section)

Two common phase changes are from solid $\leftrightarrow$ liquid (melting) and boiling); another is a change in the crystalline structure of a solid.
All such phase changes involve a change in the system's internal energy but no change in its temperature.

The increase in internal energy in boiling, for example, is represented by the breaking of bonds between molecules in the liquid state; this bond breaking allows the molecules to move farther apart in the gaseous state, with a corresponding increase in intermolecular potential energy.

Heat and work in a thermodynamic process

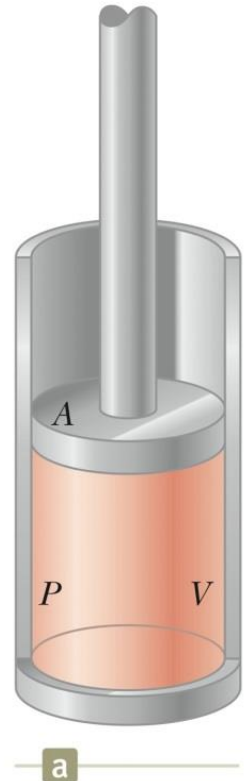
- ▶ State of a system is described by **State Variables**
 - Pressure
 - Volume
 - Temperature
 - Internal Energy
- ▶ State of a system can only be described when it is in **thermal equilibrium**.
- ▶ **Transfer-Variables**

$$\Delta K + \Delta U + \Delta E_{\text{int}} = W + Q + T_{MW} + T_{MT} + T_{ET} + T_{ER}$$

- RHS: these variables cause energy to be transferred across the system's boundaries
- Energy transfer: + in and – out of the system.

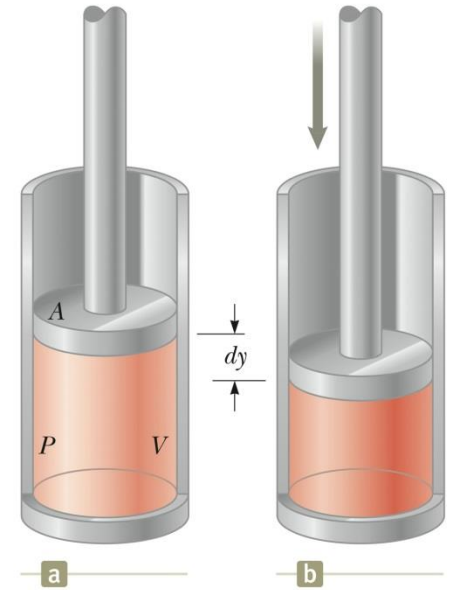
Heat and work in a thermodynamic process

- ▶ Ch 7 – Work done on a non-deformable particle or object
- ▶ Now consider the work done on a deformable system – e.g. a gas
 - At **equilibrium** – gas has a volume V
 - The gas exerts a **pressure** P on the sides of the container and the piston
 - Piston has an area, A and therefore $F = PA$
 - Upward force on the piston due to the gas is then = weight of the piston on the gas + force PA from atmospheric pressure.



Heat and work in a thermodynamic process

- ▶ If the piston is pushed down slowly – in quasi-static thermal equilibrium the whole time.
- ▶ The force on the gas downwards $\vec{F} = -F \hat{j}$
- ▶ Piston lead Displacement $d\vec{r} = dy \hat{j}$
- ▶ Work done on the gas (**system**):
$$dW = \vec{F} \cdot d\vec{r} = -F \hat{j} \cdot dy \hat{j} = -PA dy$$
$$A dy = dV \quad \dots \text{a change in volume}$$
$$dW = -PdV$$



If gas is compressed, dV is negative $\rightarrow W$ is positive.
If gas expands, dV is positive $\rightarrow W$ is negative.

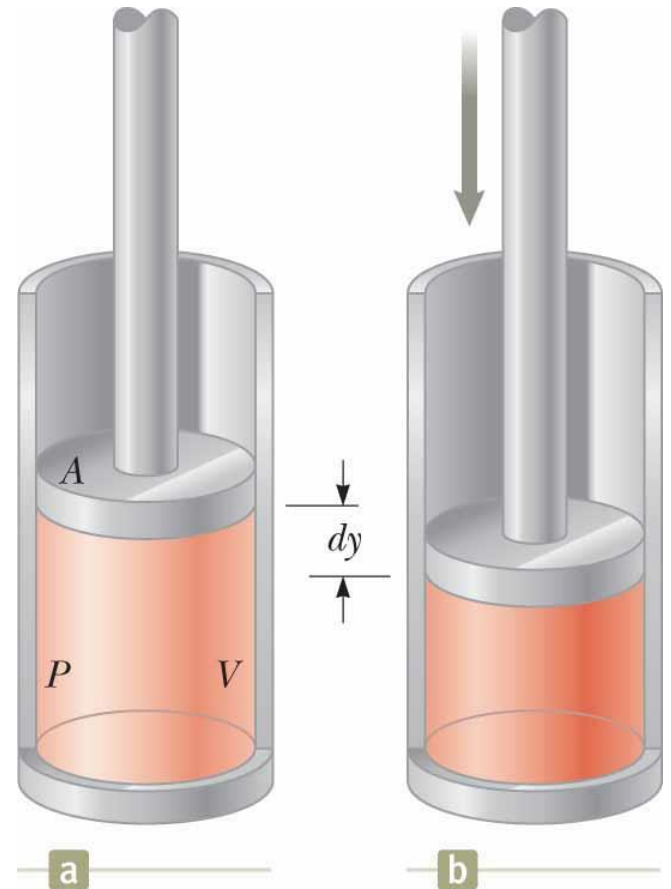
- ▶ $W = -\int PdV$, and P is usually dependent on V and T

Work Done on a Gas

$$dW = \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}} = -F\hat{\mathbf{j}} \cdot dy\hat{\mathbf{j}} = -F dy = -PA dy$$

$$dW = -PdV$$

$$W = -\int_{V_i}^{V_f} PdV$$

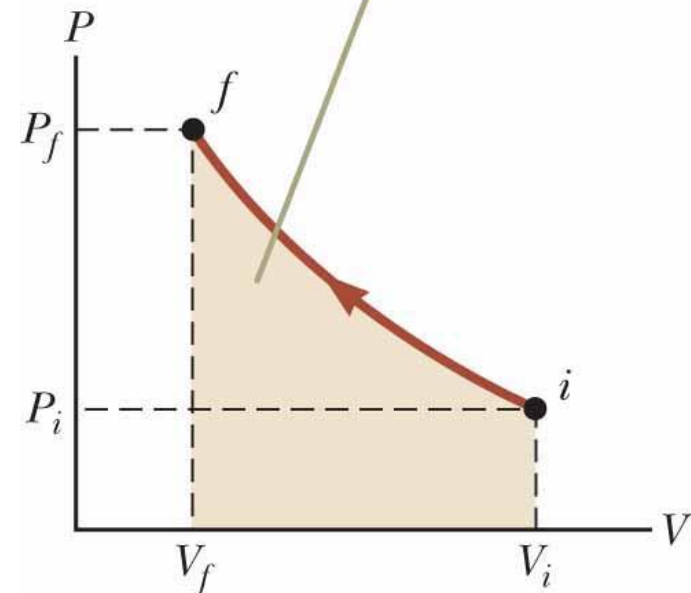


Work and PV Diagrams

$$W = -\int_{V_i}^{V_f} P dV$$

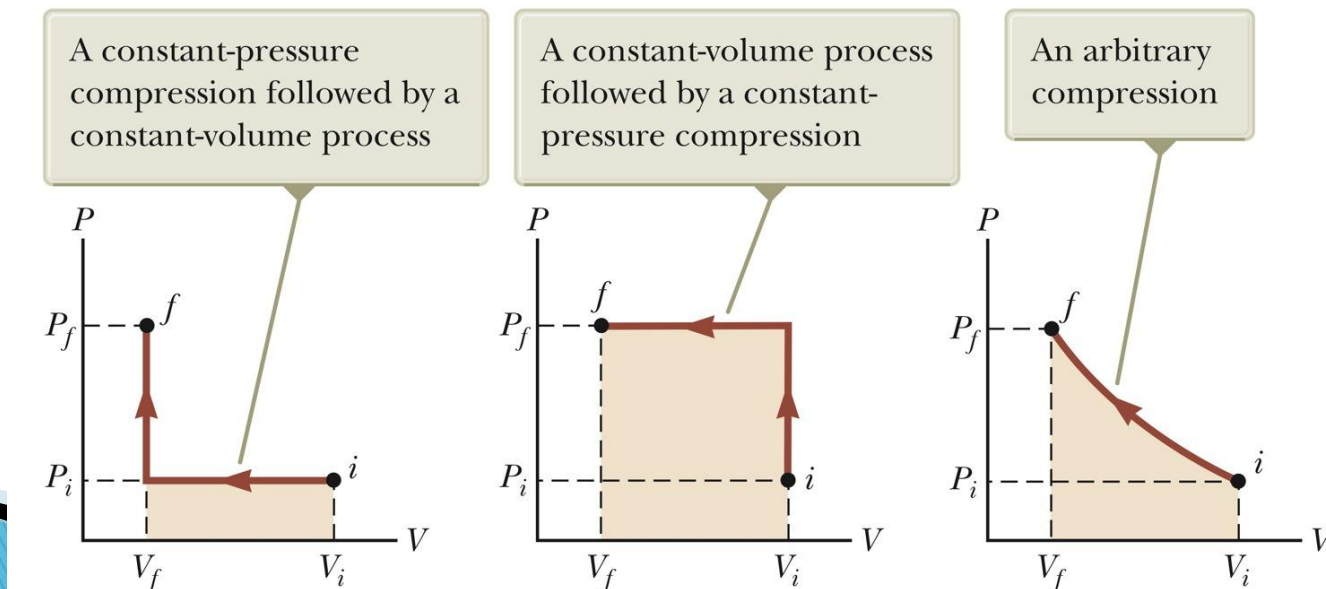
The work done on a gas in a quasi-static process that takes the gas from an initial state to a final state is the negative of the area under the curve on a PV diagram, evaluated between the initial and final states.

The work done on a gas equals the negative of the area under the PV curve. The area is negative here because the volume is decreasing, resulting in positive work.



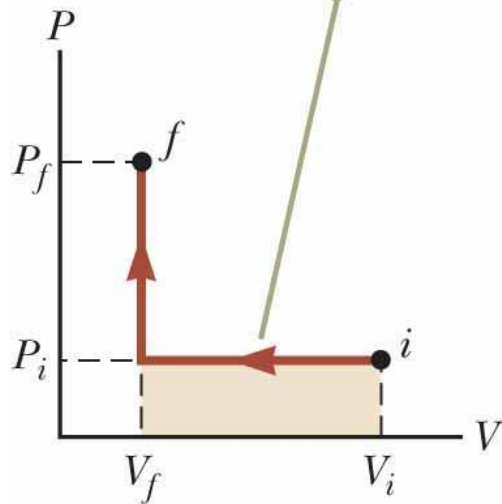
Heat and work in a thermodynamic process

- ▶ **PV diagrams**
- ▶ Lots of ways for a system to change from an initial state i to a final state f – This is represented by P - V curve
- ▶ Work done on the gas going from state i to state f is given by the **negative area under the curve**
- ▶ Work is path-dependent



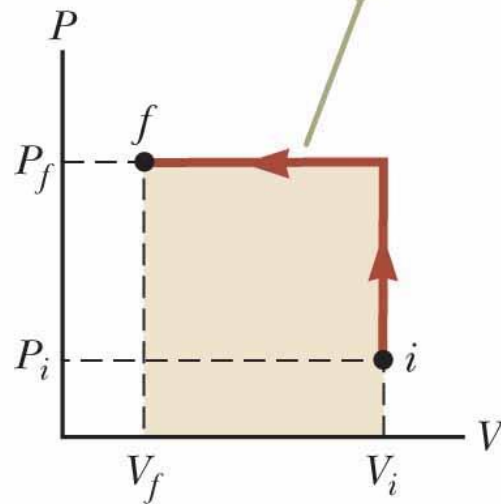
Work and PV Diagrams

A constant-pressure compression followed by a constant-volume process



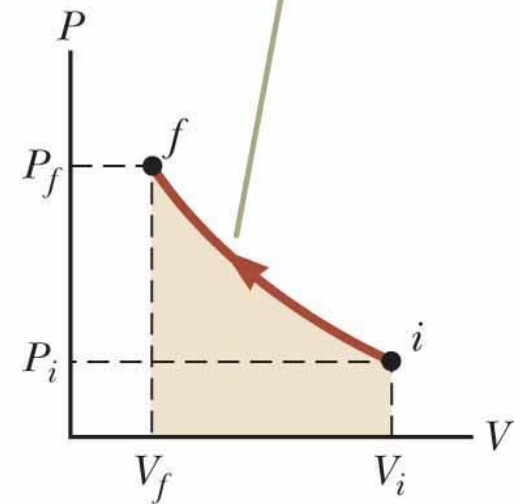
a

A constant-volume process followed by a constant-pressure compression



b

An arbitrary compression

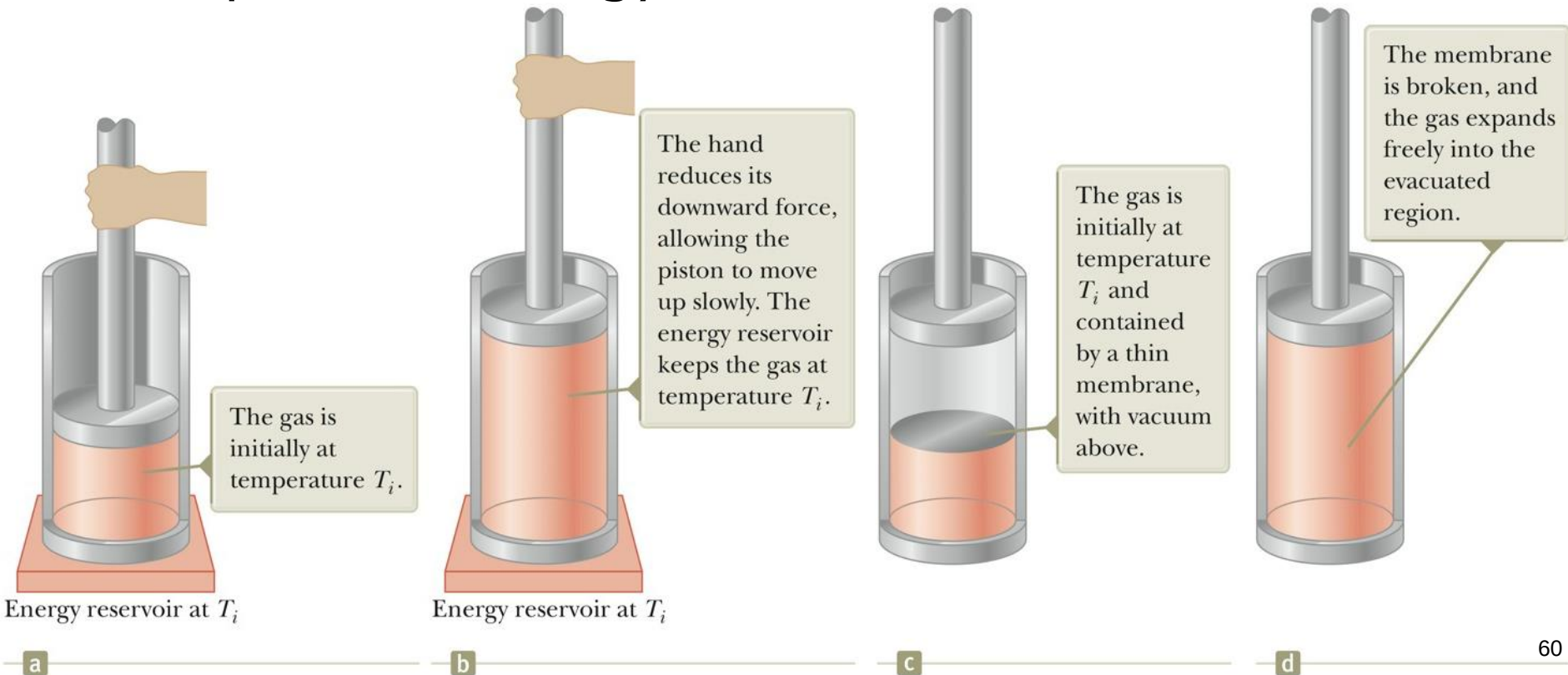


c

$$W = -P_i (V_f - V_i) \quad W = -P_f (V_f - V_i) \quad W = -\int_{V_i}^{V_f} P dV$$

Heat and work in a thermodynamic process

- ▶ Energy transferred in or out of a system in the form of heat, Q , depends on the process used.
- ▶ Ideal gas at V_i , P_i , T_i – thermally insulated on the sides of the containers.
- ▶ Except for the Energy-reservoirs in (a) and (b).

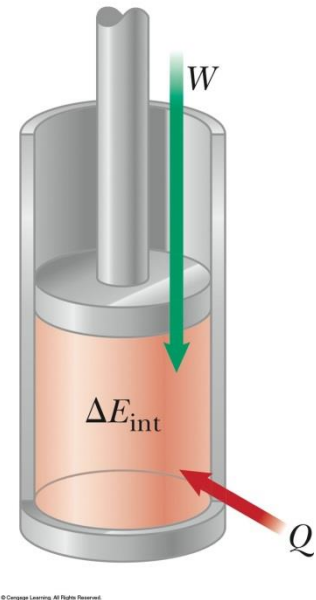


1st Law of Thermodynamics

- ▶ Ch 8 – Conservation of energy.
- ▶ When the state of a system changes from i to f , Q and W are dependent on the type of process used.
- ▶ BUT sum of $Q + W$ is always the same for the different processes.

$$Q + W = \Delta E_{\text{int}} = E_{\text{int},f} - E_{\text{int},i}$$

- ▶ First law of thermodynamics
 - ▶ For an **isolated system** (no interaction with the environment)
 - No transfer of energy in the form of heat
 - No work done on the system
- $$Q + W = 0 = \Delta E_{\text{int}}$$
- $$\therefore E_{\text{int},f} = E_{\text{int},i}$$



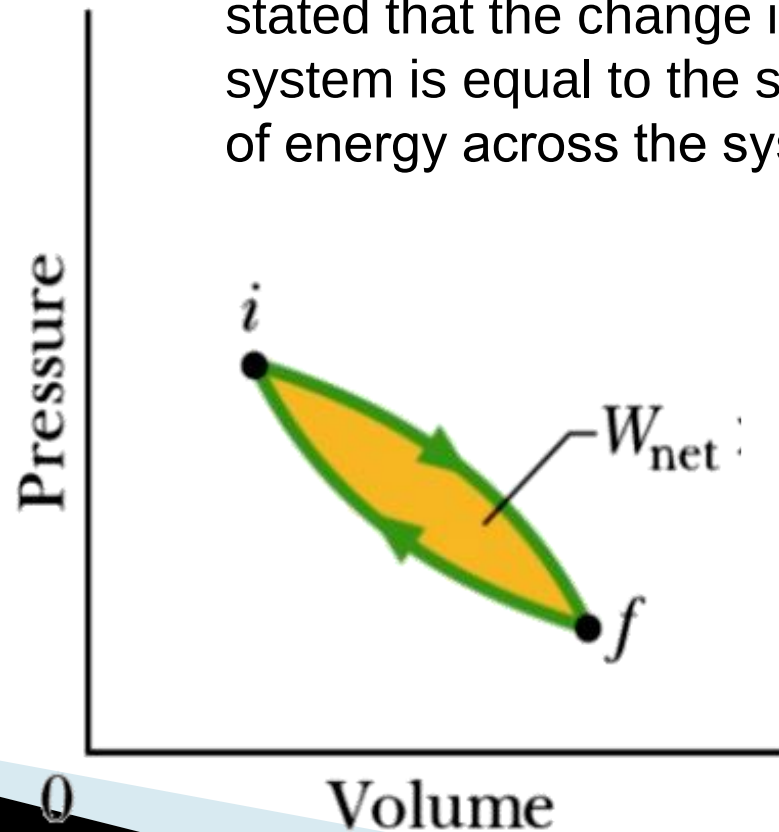
1st Law of Thermodynamics

- ▶ **Cyclic process** (process which starts and ends in the same state)
 - Change in internal energy is zero. (**State Variable!**)
 - Therefore

$$Q + W = \Delta E_{\text{int}} = 0$$

$$\therefore Q = -W$$

When we introduced the law of conservation of energy in Chapter 8, we stated that the change in the energy of a system is equal to the sum of all transfers of energy across the system's boundary



Isothermal Processes

Isothermal process: occurs at constant temperature.

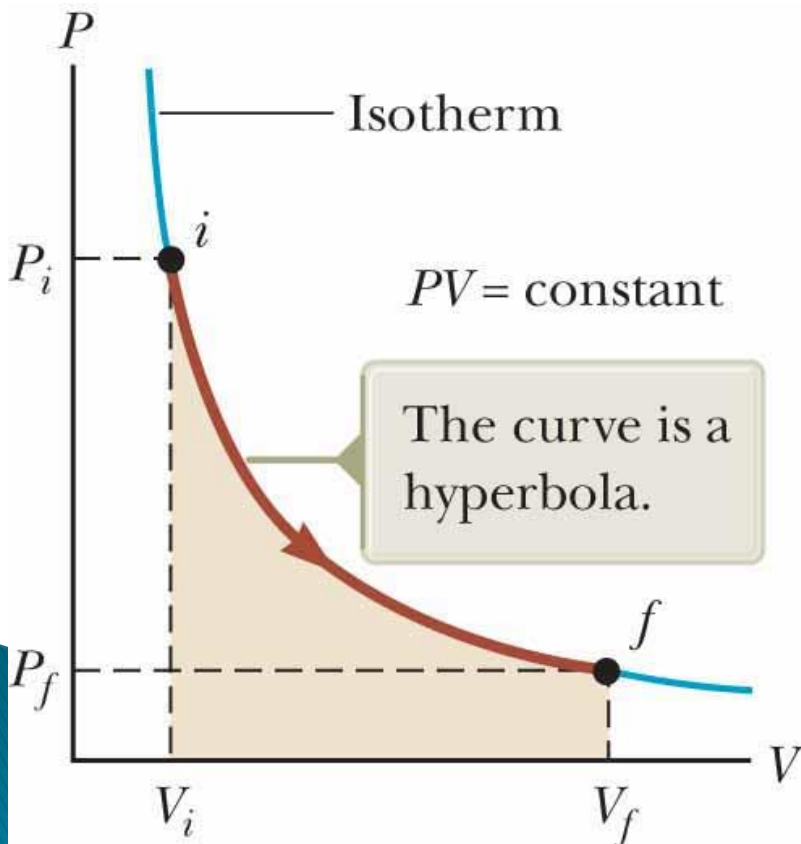
Immerse cylinder in figure in ice–water bath OR Put cylinder in contact with other constant-temperature reservoir

$$PV = nRT = \text{constant}$$

E_{int} is function of T

$$\Delta E_{\text{int}} = 0$$

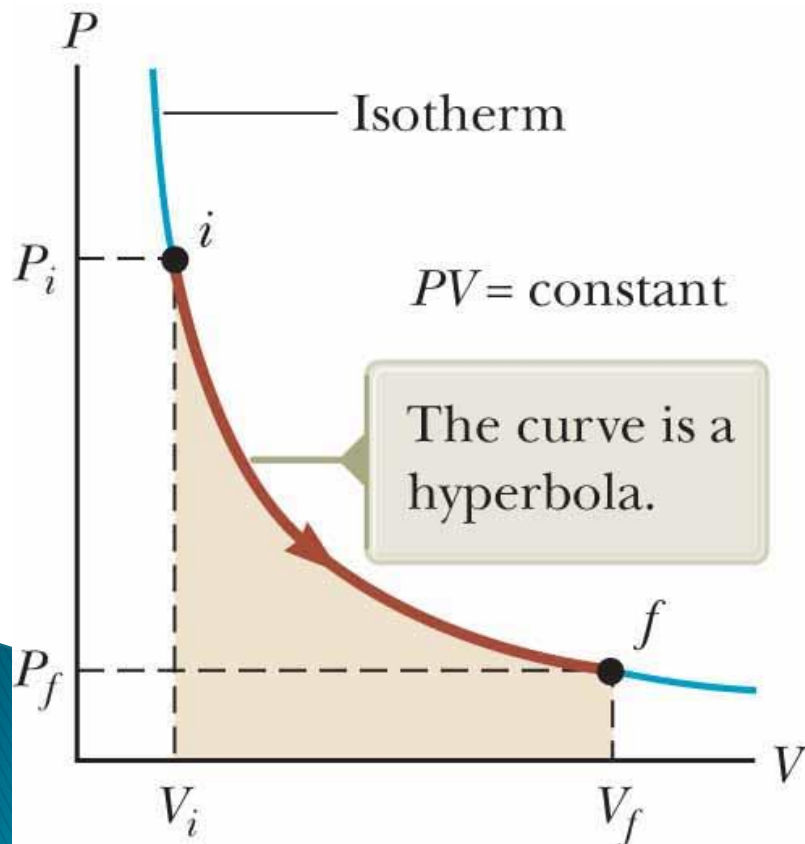
$$Q = -W$$



Isothermal Processes

$$W = -\int_{V_i}^{V_f} P dV = -\int_{V_i}^{V_f} \frac{nRT}{V} dV$$

$$W = -nRT \int_{V_i}^{V_f} \frac{dV}{V} = -nRT \ln V \Big|_{V_i}^{V_f}$$



$$W = nRT \ln \left(\frac{V_i}{V_f} \right)$$

(isothermal process)

Applications of the 1st Law

- ▶ **Adiabatic Process:** $Q = 0$,
eg. If the gas is compressed work is done (W is +ve) and the change in the internal energy is positive. $\Delta E_{\text{int}} = W$
- ▶ **Free expansion:** $W = 0$, $Q = 0$,
i.e. adiabatic process where no heat is transferred, and no work is done on or by the system. $\Delta E_{\text{int}} = 0$
- ▶ **Isovolumetric process:** $W = 0$, $\Delta E_{\text{int}} = Q$
i.e. $\Delta V = 0$, heat is absorbed or lost by the system.
$$\Delta E_{\text{int}} = Q + W$$
- ▶ **Isobaric process:** $W = -P(V_f - V_i)$,
eg. The system has a constant pressure throughout the process (Fig 19.6 a and b).

Isobaric Processes

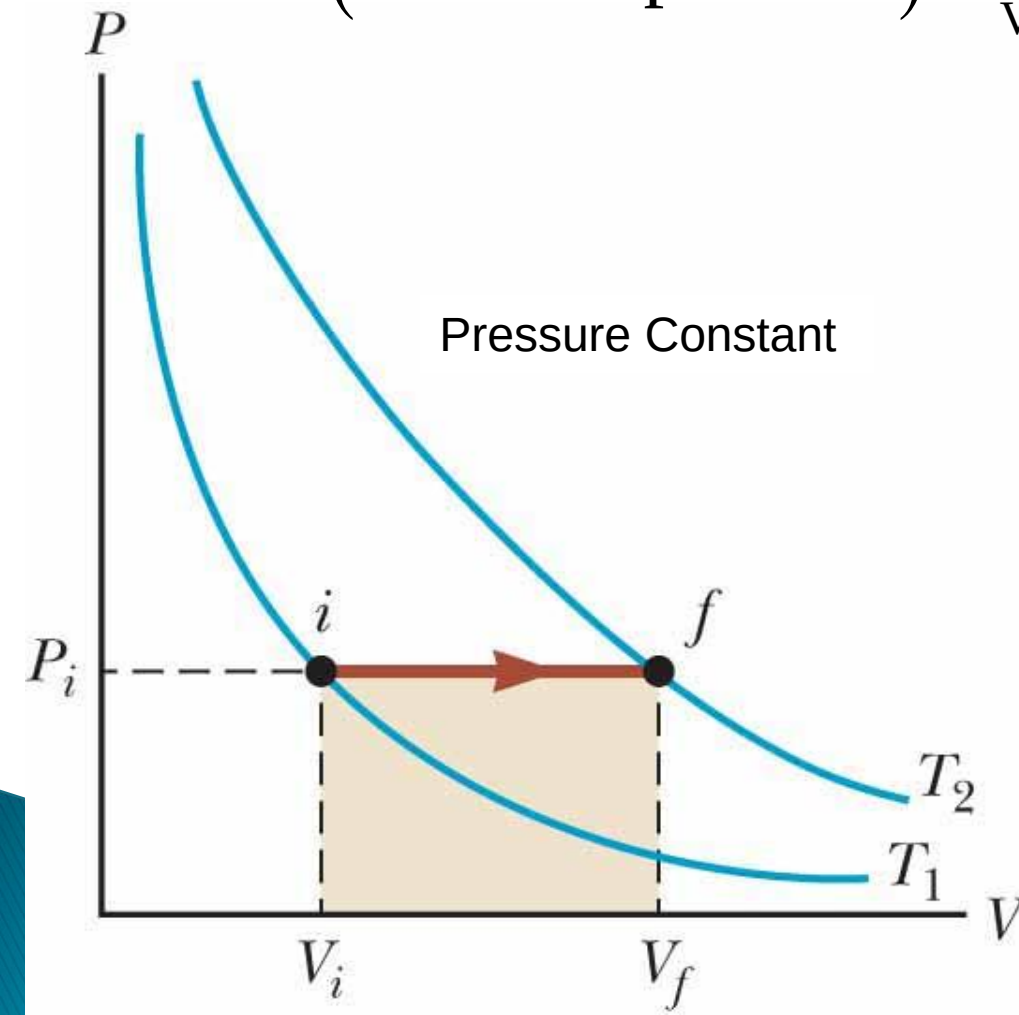
$$W = -P(V_f - V_i)$$

(isobaric process)

Could be established by allowing piston to move freely: always in equilibrium between net force from gas pushing upward and weight of piston plus force due to atmospheric pressure pushing downward

Isobaric process: horizontal line on PV diag/m.

Values of heat and work both usually nonzero



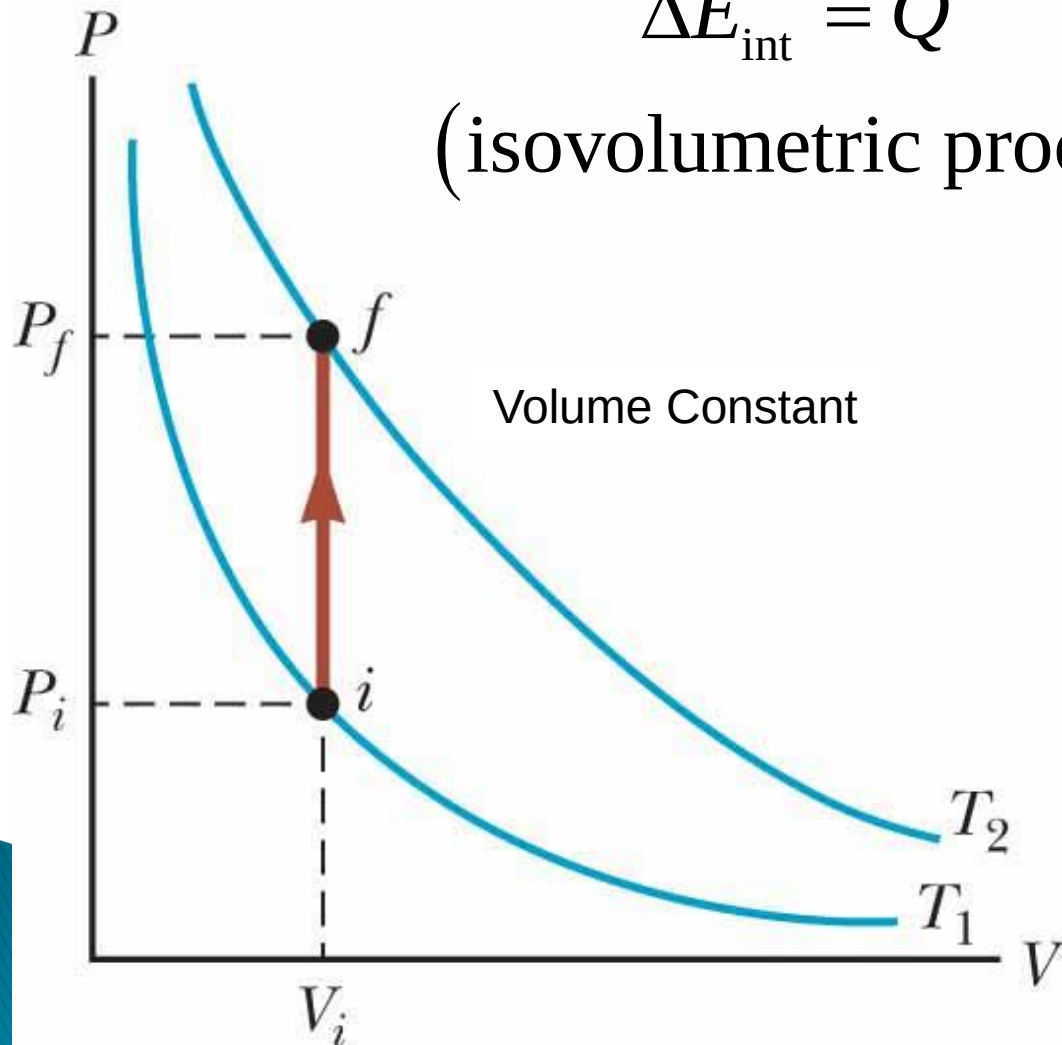
Isovolumetric Processes

Takes place at constant volume (*isochoric*) Could be established by clamping piston at fixed position. Isovolumetric process: vertical line on PV diagram

Volume of gas does not change: Work = 0

$$\Delta E_{\text{int}} = Q$$

(isovolumetric process)



Adiabatic Processes

When $Q = 0$ $\Delta E_{\text{int}} = W$
(adiabatic process)



Applications of the 1st Law

- ▶ **Isothermal process:** $\Delta T = 0$, $\Delta E_{\text{int}} = 0$

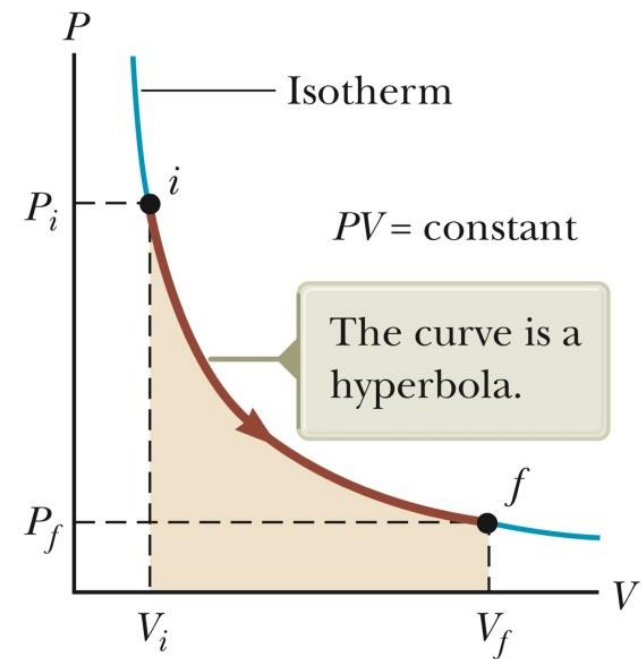
$$\therefore Q = -W$$

- ▶
$$W = -\int_{V_i}^{V_f} P dV = -\int_{V_i}^{V_f} \frac{nRT}{V} dV$$

$$W = -nRT \int_{V_i}^{V_f} \frac{dV}{V} = -nRT \ln V \Big|_{V_i}^{V_f}$$

$$= nRT \ln \frac{V_i}{V_f}$$

Because the gas expands, $V_f > V_i$ and the value for the work done on the gas is negative as we expect.
If the gas is compressed, then $V_f < V_i$ and the work done on the gas is positive.



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- eg. Any energy that enters the system by heat is transferred out of the system by work; as a result, no change in the internal energy of the system occurs (external force does negative work).

$$\Delta E_{\text{int}} = 0 \quad \therefore Q = -W$$

Applications of the 1st Law

- ▶ **Adiabatic Process:** $Q = 0$,
eg. If the gas is compressed work is done (W is +ve) and the change in the internal energy is positive. $\Delta E_{\text{int}} = W$
- ▶ **Free expansion:** $W = 0$, $Q = 0$,
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eg. The system has a constant pressure throughout the process (Fig 19.6 a and b).

Quick Quiz 19.3

In the last three columns of the following table, fill in the boxes with the correct signs (+, −, or 0) for Q , W , and ΔE_{int} . For each situation, the system to be considered is identified.

Situation	System	Q	W	ΔE_{int}
Rapidly pumping up a bicycle tire	Air in the pump			
Pan of room-temperature water sitting on a hot stove	Water in the pan			
Air quickly leaking out of a balloon	Air originally in the balloon			

Quick Quiz 19.3

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$$\Delta E_{\text{int}} = Q + W$$

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Quick Quiz 19.3

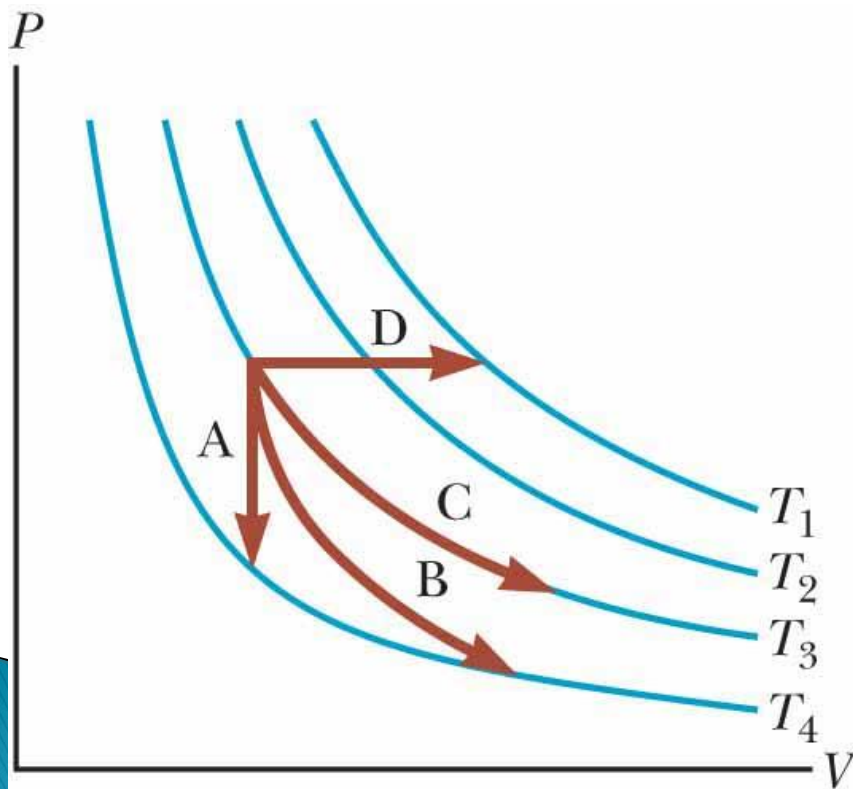
In the last three columns of the following table, fill in the boxes with the correct signs (+, −, or 0) for Q , W , and ΔE_{int} . For each situation, the system to be considered is identified.

$$\Delta E_{\text{int}} = Q + W$$

Situation	System	Q	W	ΔE_{int}
Rapidly pumping up a bicycle tire	Air in the pump	0	+	+
Pan of room-temperature water sitting on a hot stove	Water in the pan	+	0	+
Air quickly leaking out of a balloon	Air originally in the balloon	0	−	−

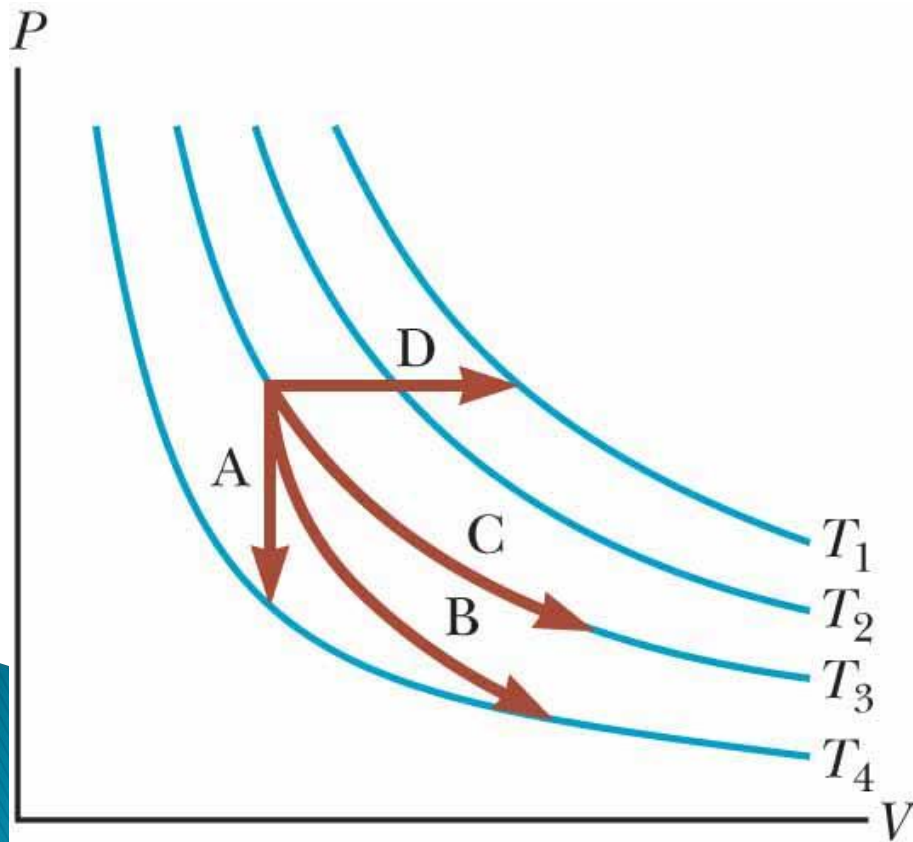
Quick Quiz 19.4

Characterize the paths in the figure as isobaric, isovolumetric, isothermal, or adiabatic. For path B, $Q = 0$. The blue curves are isotherms.



Quick Quiz 19.4

Characterize the paths in the figure as isobaric, isovolumetric, isothermal, or adiabatic. For path B, $Q = 0$. The blue curves are isotherms.

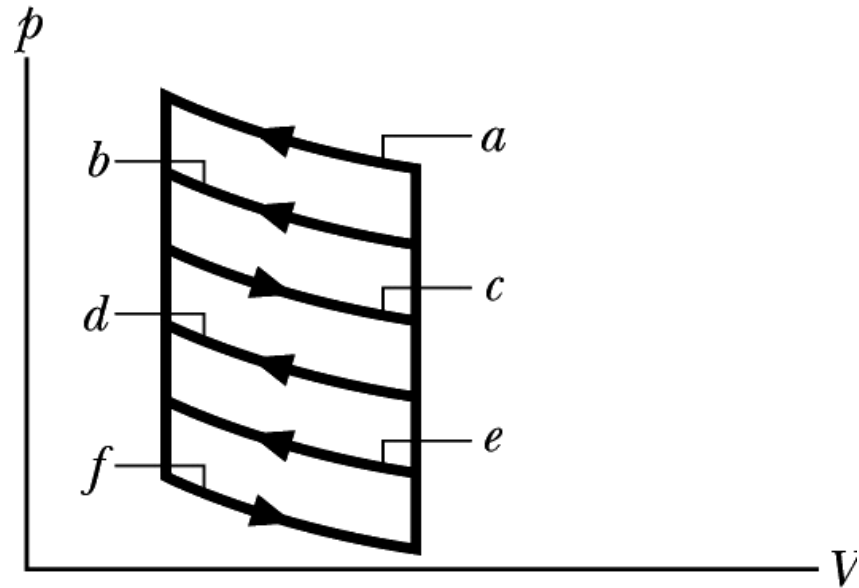


Path A is isovolumetric, path B is adiabatic, path C is isothermal, and path D is isobaric.

Example

- ▶ The P - V diagram here shows 6 curved paths (connected by vertical paths) that can be followed by a gas. Which 2 of the curved paths should be part of a closed cycle (those curved paths plus connecting vertical paths) if the net work done on the gas during the cycle is to be at its
- ▶ (a) maximum negative value?
- ▶ (b) maximum positive value?

$$W = -P(V_f - V_i)$$



Example

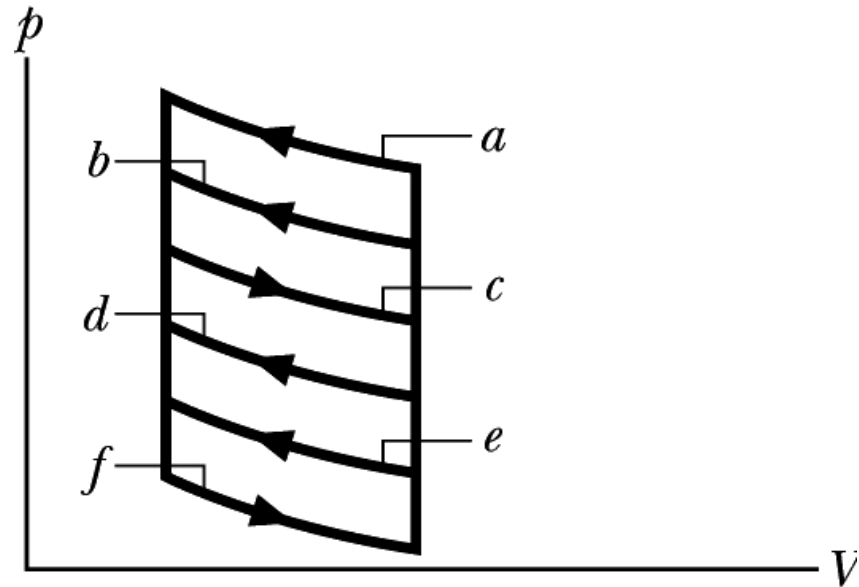
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 - ▶ (a) maximum negative value?
 - ▶ (b) maximum positive value?

$$W = -P(V_f - V_i)$$

Work is neg area under curve

For (a), to be neg and max, we need a pos $(V_f - V_i)$; approaching from top to bottom curves.

1st bottom leg with small W is (e).

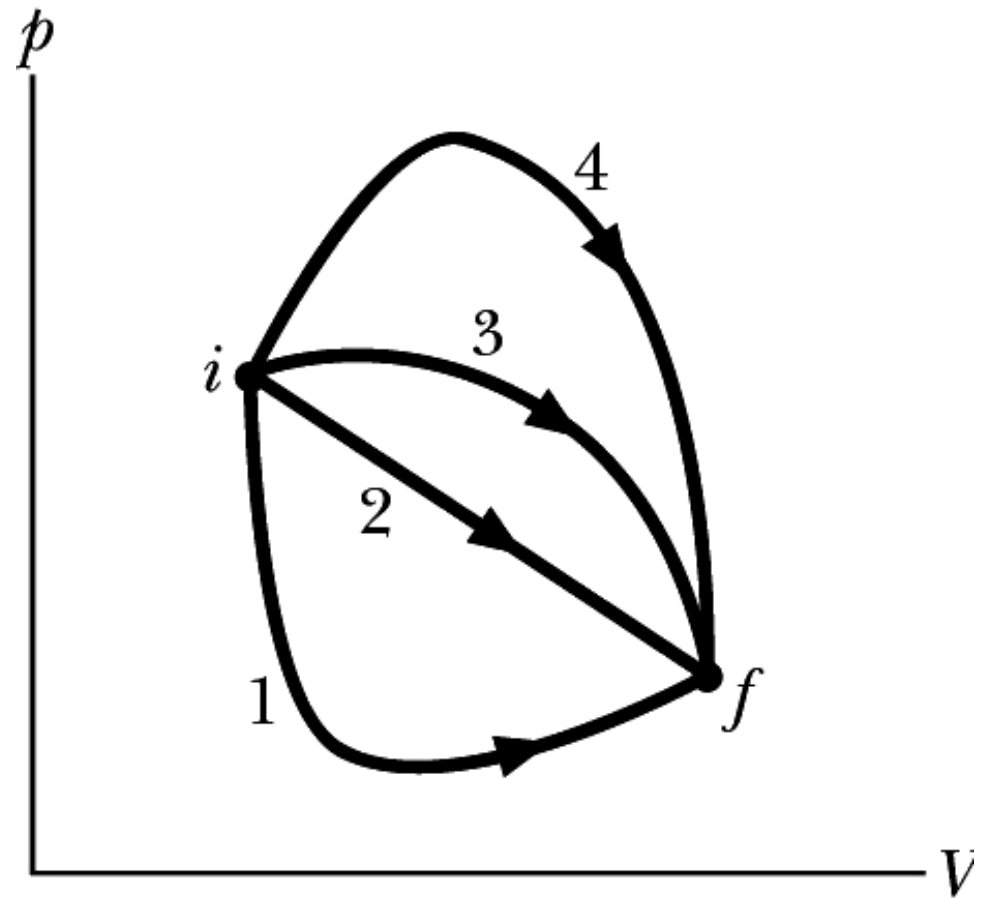


(a) c - e max area enclosed by clockwise cycle.

(b) a - f max area enclosed by anti-clockwise cycle.

Example

- ▶ The figure shows 4 paths on the P - V diagram along which a gas can be taken from state i to f . Rank the paths according to (a) the change ΔE_{int} of the gas, (b) the work done by the gas, and (c) the magnitude of the energy transferred as heat Q between gas and the environment, greatest first.



Example

- The figure shows 4 paths on the P - V diagram along which a gas can be taken from state i to f . Rank the paths according to (a) the change ΔE_{int} of the gas, (b) the work done by the gas, and (c) the magnitude of the energy transferred as heat Q between gas and the environment, greatest first.

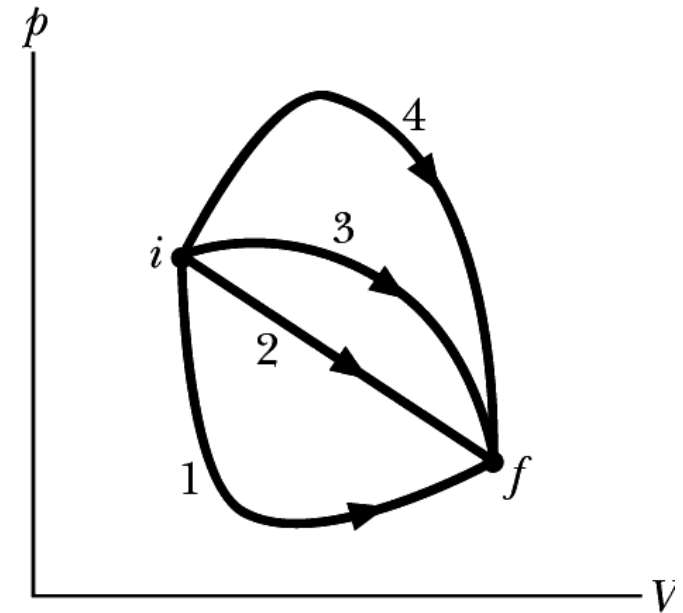
(a) All the same

(b) 4,3,2,1

Area under the curve

(c) 4,3,2,1

$Q + W$ must be constant and W is negative in all cases
 $\text{constant} = \Delta E_{\text{int}} = Q + W$



$$W = -P(V_f - V_i)$$

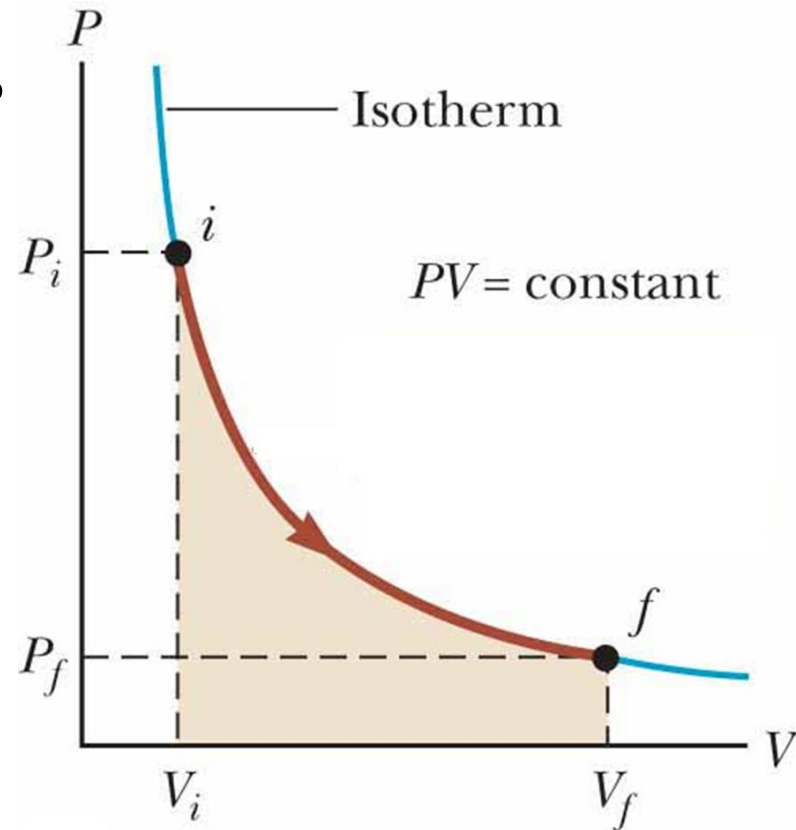
Example 19.5: An Isothermal Expansion

A 1.0-mol sample of an ideal gas is kept at 0.0°C during an expansion from 3.0 L to 10.0 L.

(A) How much work is done on the gas during the expansion?

$$W = nRT \ln \left(\frac{V_i}{V_f} \right)$$

$$= (1.0 \text{ mol})(8.31 \text{ J/mol} \cdot \text{K})(273 \text{ K}) \ln \left(\frac{3.0 \text{ L}}{10.0 \text{ L}} \right) = -2.7 \times 10^3 \text{ J}$$



Example 19.5: An Isothermal Expansion

(B) How much energy transfer by heat occurs between the gas and its surroundings in this process?

$$\Delta E_{\text{int}} = Q + W$$

Isothermal process $0 = Q + W$

$$Q = -W = 2.7 \times 10^3 \text{ J}$$

Example 19.5: An Isothermal Expansion

(C) If the gas is returned to the original volume by means of an isobaric process, how much work is done on the gas?

$$\begin{aligned} W &= -P(V_f - V_i) = -\frac{nRT_i}{V_i}(V_f - V_i) \\ &= -\frac{(1.0 \text{ mol})(8.31 \text{ J/mol} \cdot \text{K})(273 \text{ K})}{10.0 \times 10^{-3} \text{ m}^3} \times \\ &\quad (3.0 \times 10^{-3} \text{ m}^3 - 10.0 \times 10^{-3} \text{ m}^3) \\ &= \boxed{1.6 \times 10^3 \text{ J}} \end{aligned}$$

Example 19.6: Boiling Water

Suppose 1.00 g of water vaporizes isobarically at atmospheric pressure ($1.013 \times 10^5 \text{ Pa}$). Its volume in the liquid state is $V_i = V_{\text{liquid}} = 1.00 \text{ cm}^3$, and its volume in the vapor state is $V_f = V_{\text{vapor}} = 1\,671 \text{ cm}^3$. Find the work done in the expansion and the change in internal energy of the system. Ignore any mixing of the steam and the surrounding air; imagine that the steam simply pushes the surrounding air out of the way.

Example 19.6: Boiling Water

$$W = -P(V_f - V_i) \quad \text{Find the work done in the expansion - Isobaric}$$
$$= -(1.013 \times 10^5 \text{ Pa})(1671 \times 10^{-6} \text{ m}^3 - 1.00 \times 10^{-6} \text{ m}^3)$$
$$= \boxed{-169 \text{ J}} \quad \text{This energy went to the environment as Work done by steam on the surrounding atmosphere.}$$

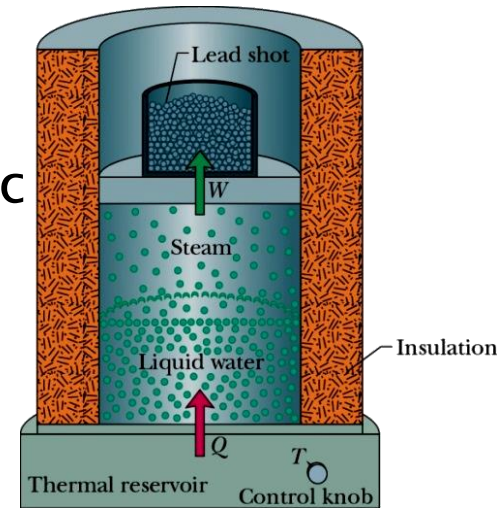
$$Q = L_v \Delta m_s = L_v m_s = (2.26 \times 10^6 \text{ J/kg})(1.00 \times 10^{-3} \text{ kg})$$
$$= \boxed{2260 \text{ J}} \quad \text{Find the heat transfer to make steam}$$

$$\Delta E_{\text{int}} = Q + W = 2260 \text{ J} + (-169 \text{ J}) = \boxed{2.09 \text{ kJ}}$$

From 1st Law of Thermodynamics

Problem

Let 1.00 kg of liquid water be at 100 °C be converted to steam at 100 °C by boiling at standard atmospheric pressure ($1.00 \text{ atm} = 1.01 \times 10^5 \text{ Pa}$) in an arrangement shown. The volume of the water changes from $1.00 \times 10^{-3} \text{ m}^3$ as a liquid to 1.671 m^3 as steam.



- How much work is done on the system during this process?
- How much energy is transferred as heat during the process?
- What is the change in the system's internal energy?

- Work done: Volume increases and pressure is constant, work done - negative:

$$\begin{aligned} W &= -\int_{V_i}^{V_f} p dV = -p \int_{V_i}^{V_f} dV \\ &= -p(V_f - V_i) = -(1.01 \times 10^5 \text{ Pa})(1.671 - 1 \times 10^{-3} \text{ m}^3) \\ &= -169 \times 10^3 \text{ J} = -169 \text{ kJ} \end{aligned}$$

- Heat transferred during phase change:

$$\begin{aligned} Q &= mL_v = (1.00 \text{ kg})(2256 \text{ kJ/kg}) \\ &= 2256 \times 10^3 \text{ J} = 2256 \text{ kJ} \end{aligned}$$

- Change in internal energy:

$$\begin{aligned} \Delta E_{\text{int}} &= Q + W = 2256 - 169 \\ &= 2087 \text{ kJ} = 2.09 \text{ MJ} \end{aligned}$$

Example 19.7: Heating a Solid

A 1.0-kg bar of copper is heated at atmospheric pressure so that its temperature increases from 20°C to 50°C.

(A) What is the work done on the copper bar by the surrounding atmosphere?

$$W = -P\Delta V$$

$$W = -P(\beta V_i \Delta T) = -P(3\alpha V_i \Delta T) = -3\alpha P V_i \Delta T$$

$$W = -3\alpha P \left(\frac{m}{\rho} \right) \Delta T$$

$$W = -3 \left[1.7 \times 10^{-5} (\text{°C})^{-1} \right] (1.013 \times 10^5 \text{ N/m}^2) \left(\frac{1.0 \text{ kg}}{8.92 \times 10^3 \text{ kg/m}^3} \right) (50\text{°C} - 20\text{°C})$$

$$= \boxed{-1.7 \times 10^{-2} \text{ J}}$$

Surrounding atmosphere “allows” the Cu bar to expand into it.
That is. The Cu bar expands against the atmosphere

Example 19.7: Heating a Solid

A 1.0-kg bar of copper is heated at atmospheric pressure so that its temperature increases from 20°C to 50°C.

(B) How much energy is transferred to the copper by heat?

(C) What is the increase in the internal energy of the copper bar?

$$Q = C_{Cu} \Delta T = 1.16 \times 10^4 \text{ J}$$

$$\Delta E_{\text{int}} = W + Q$$

Energy transfer mechanisms – not examinable

- ▶ two other energy transfer methods often related to temperature changes: convection (a form of matter transfer TMT) and electromagnetic radiation TE
- ▶ The process of energy transfer by heat (Q in Eq. 8.2) can also be called conduction or thermal conduction. In this process, the transfer can be represented on an atomic scale as an exchange of kinetic energy between microscopic particles—molecules, atoms, and free electrons—in which less-energetic particles gain energy in collisions with more-energetic particles.
- ▶ The rate of thermal conduction through a material depends on the properties of the material. For example, it is possible to hold a piece of asbestos in a flame indefinitely, which implies that very little energy is conducted through the asbestos. In general, metals are good thermal conductors and materials such as asbestos, cork, paper, and fiberglass are poor conductors.

Energy transfer mechanisms – not examinable

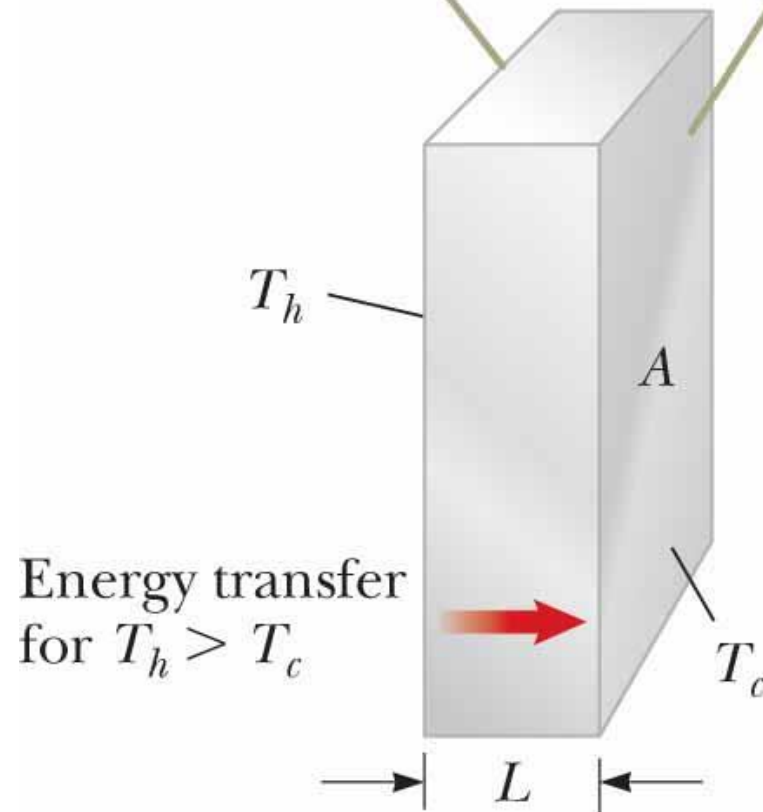
- ▶ Experimentally, it is found that energy Q transfers in a time interval Δt from the hotter face to the colder one. The energy Q that transfers is found to be proportional to the cross-sectional area, the temperature difference $\Delta T = T_h - T_c$, and the time interval, and inversely proportional to the thickness:

Thermal Conduction

$$Q = kA \frac{\Delta T}{\Delta x} \Delta t$$

$$P = kA \left| \frac{dT}{dx} \right|$$

The opposite faces are at different temperatures where $T_h > T_c$.



Thermal Conductivities

TABLE 19.3 Thermal
Conductivities

Substance	Thermal Conductivity (W/m · °C)
<i>Metals (at 25°C)</i>	
Aluminum	238
Copper	397
Gold	314
Iron	79.5
Lead	34.7
Silver	427
<i>Nonmetals (approximate values)</i>	
Asbestos	0.08
Concrete	0.8
Diamond	2 300
Glass	0.8
Ice	2
Rubber	0.2
Water	0.6
Wood	0.08
<i>Gases (at 20°C)</i>	
Air	0.023 4
Helium	0.138
Hydrogen	0.172
Nitrogen	0.023 4
Oxygen	0.023 8

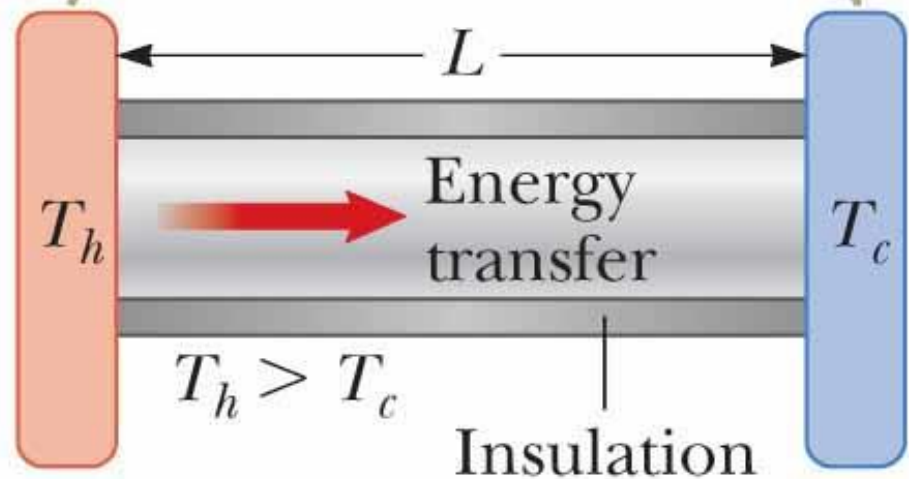
Thermal Conduction

$$\left| \frac{dT}{dx} \right| = \frac{T_h - T_c}{L}$$

$$P = kA \left(\frac{T_h - T_c}{L} \right)$$

$$P = \frac{A(T_h - T_c)}{\sum_i (L_i / k_i)}$$

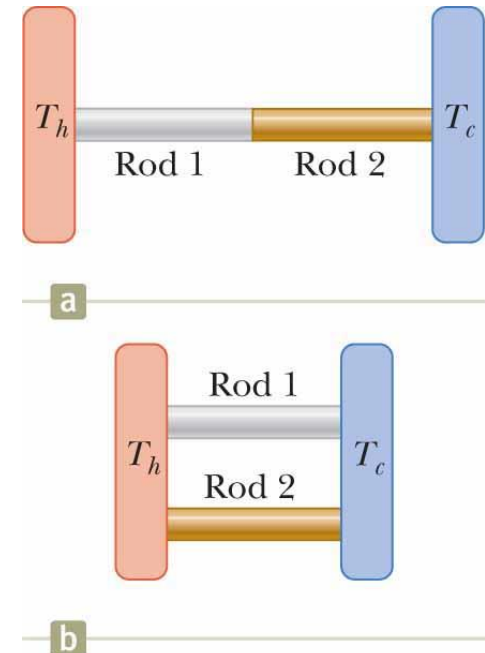
The opposite ends of the rod are in thermal contact with energy reservoirs at different temperatures.



Quick Quiz 19.5

You have two rods of the same length and diameter, but they are formed from different materials. The rods are used to connect two regions at different temperatures so that energy transfers through the rods by heat. They can be connected in series as the top figure or in parallel as in the bottom figure. In which case is the rate of energy transfer by heat larger?

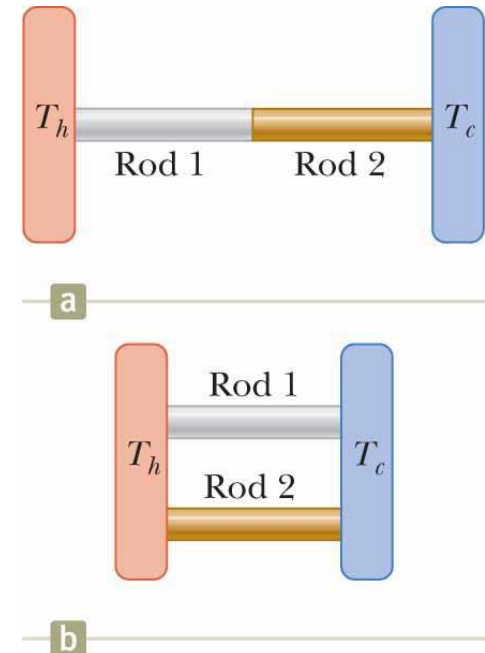
- (a) The rate is larger when the rods are in series.
- (b) The rate is larger when the rods are in parallel.
- (c) The rate is the same in both cases.



Quick Quiz 19.5

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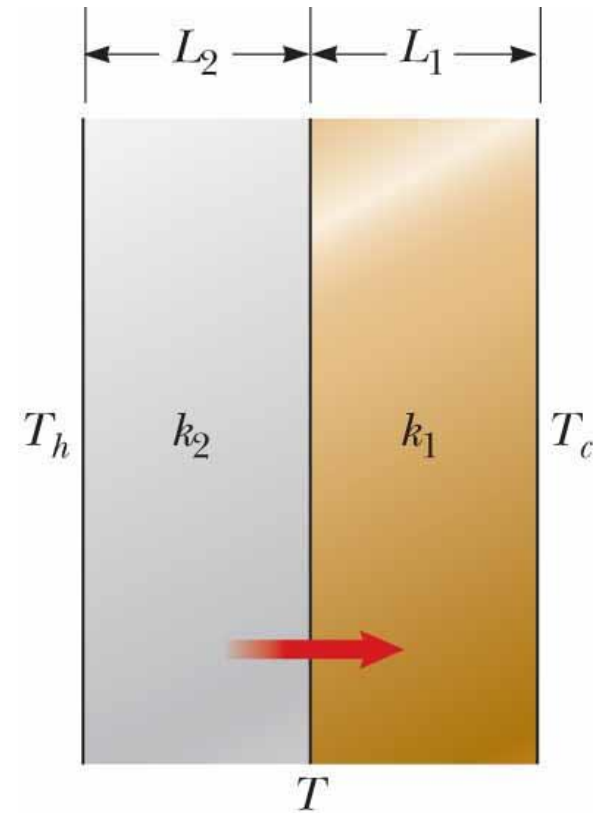


Example 19.7:

Energy Transfer Through Two Slabs

Two slabs of thickness L_1 and L_2 and thermal conductivities k_1 and k_2 are in thermal contact with each other as shown in the figure. The temperatures of their outer surfaces are T_c and T_h , respectively, and $T_h > T_c$.

Determine the temperature at the interface and the rate of energy transfer by conduction through an area A of the slabs in the steady-state condition.



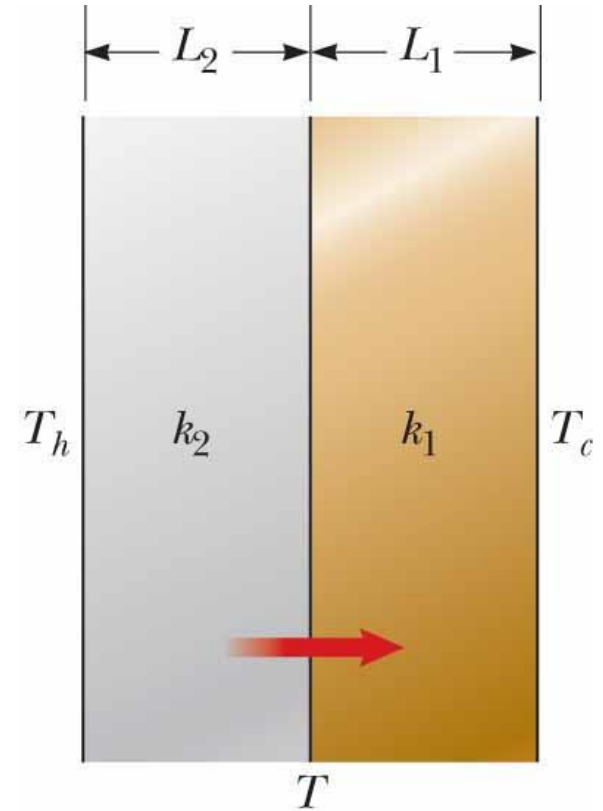
Example 19.7:

Energy Transfer Through Two Slabs

$$P_1 = k_1 A \left(\frac{T - T_c}{L_1} \right)$$

$$P_2 = k_2 A \left(\frac{T_h - T}{L_2} \right)$$

$$k_1 A \left(\frac{T - T_c}{L_1} \right) = k_2 A \left(\frac{T_h - T}{L_2} \right)$$



$$T = \frac{k_1 L_2 T_c + k_2 L_1 T_h}{k_1 L_2 + k_2 L_1}$$

$$P = \frac{A(T_h - T_c)}{(L_1/k_1) + (L_2/k_2)}$$

Example 19.7: Energy Transfer Through Two Slabs

Suppose you are building an insulated container with two layers of insulation and the rate of energy transfer turns out to be too high. You have enough room to increase the thickness of one of the two layers by 20%. How would you decide which layer to choose?

$$P = \frac{A(T_h - T_c)}{(L_1/k_1) + (L_2/k_2)}$$

Home Insulation

$$P = \frac{A(T_h - T_c)}{\sum_i (L_i / k_i)}$$

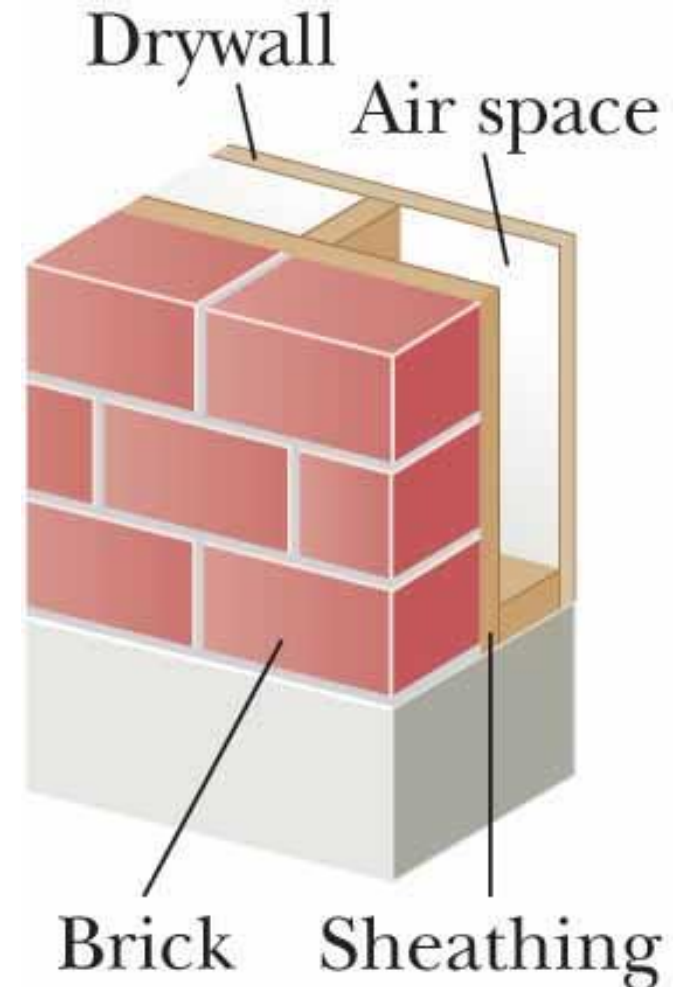
$$P = \frac{A(T_h - T_c)}{\sum_i R_i}$$

TABLE 19.4 R-Values for Some Common Building Materials

Material	<i>R</i> -value (ft ² · °F · h/Btu)
Hardwood siding (1 in. thick)	0.91
Wood shingles (lapped)	0.87
Brick (4 in. thick)	4.00
Concrete block (filled cores)	1.93
Fiberglass insulation (3.5 in. thick)	10.90
Fiberglass insulation (6 in. thick)	18.80
Fiberglass board (1 in. thick)	4.35
Cellulose fiber (1 in. thick)	3.70
Flat glass (0.125 in. thick)	0.89
Insulating glass (0.25-in. space)	1.54
Air space (3.5 in. thick)	1.01
Stagnant air layer	0.17
Drywall (0.5 in. thick)	0.45
Sheathing (0.5 in. thick)	1.32

Example 19.8: The R -Value of a Typical Wall

Calculate the total R -value for a wall constructed as shown in the figure. Starting outside the house (toward the front in the figure) and moving inward, the wall consists of 4 in. of brick, 0.5 in. of sheathing, an air space 3.5 in. thick, and 0.5 in. of drywall.



Example 19.8: The R -Value of a Typical Wall

$$R_1 \text{ (outside stagnant air layer)} = 0.17 \text{ ft}^2 \cdot ^\circ\text{F} \cdot \text{h/Btu}$$

$$R_2 \text{ (brick)} = 4.00 \text{ ft}^2 \cdot ^\circ\text{F} \cdot \text{h/Btu}$$

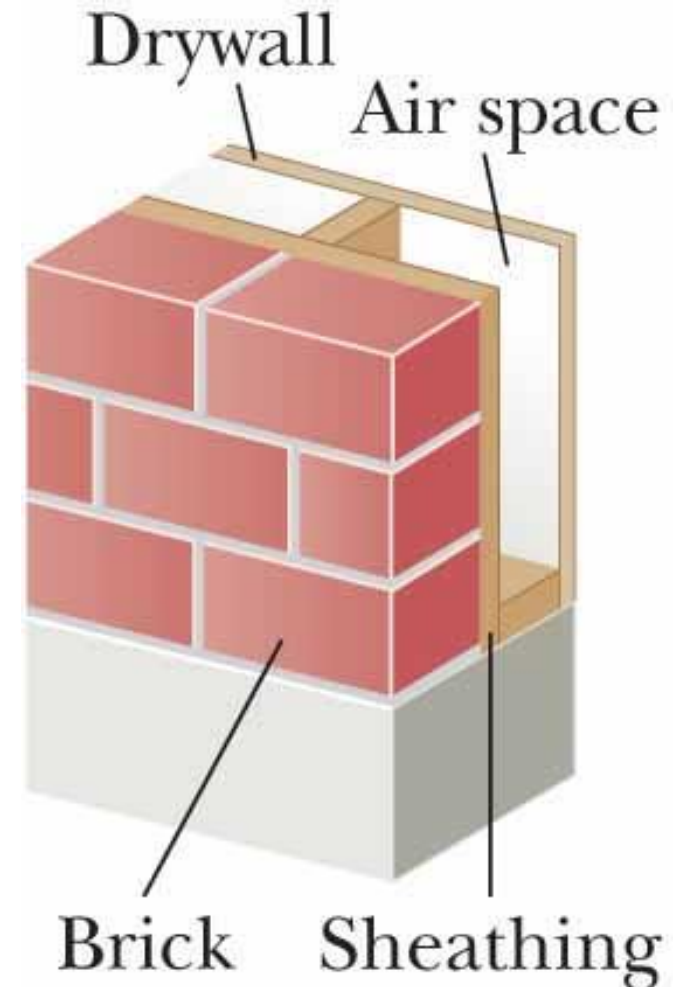
$$R_3 \text{ (sheathing)} = 1.32 \text{ ft}^2 \cdot ^\circ\text{F} \cdot \text{h/Btu}$$

$$R_4 \text{ (air space)} = 1.01 \text{ ft}^2 \cdot ^\circ\text{F} \cdot \text{h/Btu}$$

$$R_5 \text{ (drywall)} = 0.45 \text{ ft}^2 \cdot ^\circ\text{F} \cdot \text{h/Btu}$$

$$R_6 \text{ (inside stagnant air layer)} \\ = 0.17 \text{ ft}^2 \cdot ^\circ\text{F} \cdot \text{h/Btu}$$

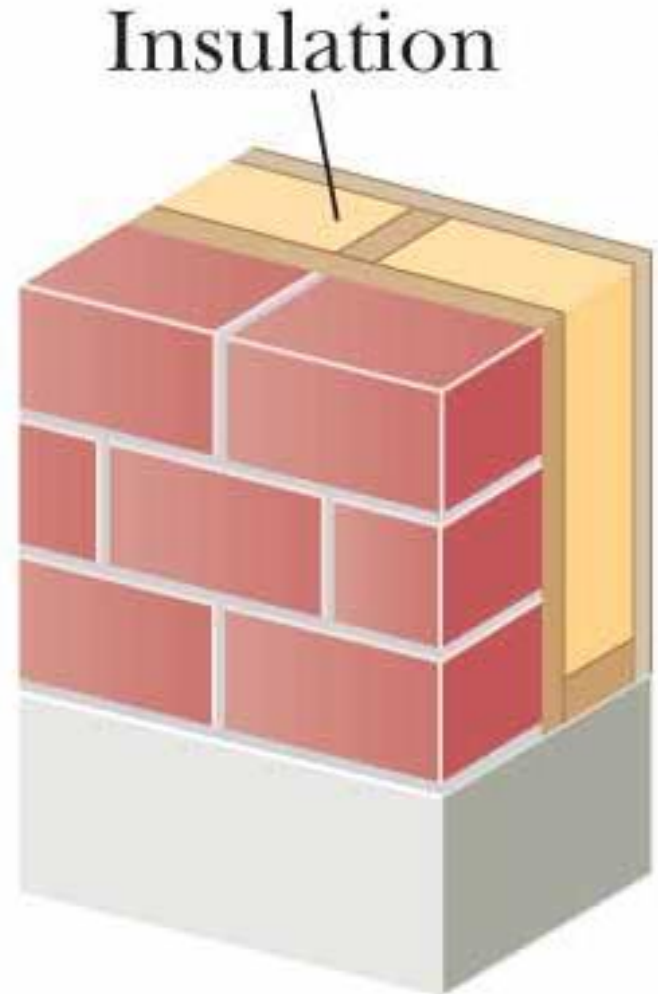
$$R_{\text{total}} = R_1 + R_2 + R_3 + R_4 + R_5 + R_6 \\ = \boxed{7.12 \text{ ft}^2 \cdot \text{F} \cdot \text{h/Btu}}$$



Example 19.8: The R -Value of a Typical Wall

Suppose you are not happy with this total R -value for the wall. You cannot change the overall structure, but you can fill the air space as in the figure. To maximize the total R -value, what material should you choose to fill the air space?

$$\begin{aligned} R_{\text{total}} &= 7.12 \text{ ft}^2 \cdot \text{F} \cdot \text{h/Btu} \\ &\quad + 9.89 \text{ ft}^2 \cdot \text{F} \cdot \text{h/Btu} \\ &= 17.01 \text{ ft}^2 \cdot \text{F} \cdot \text{h/Btu} \end{aligned}$$



Convection



Radiation

$$P = \sigma AeT^4$$



Radiation

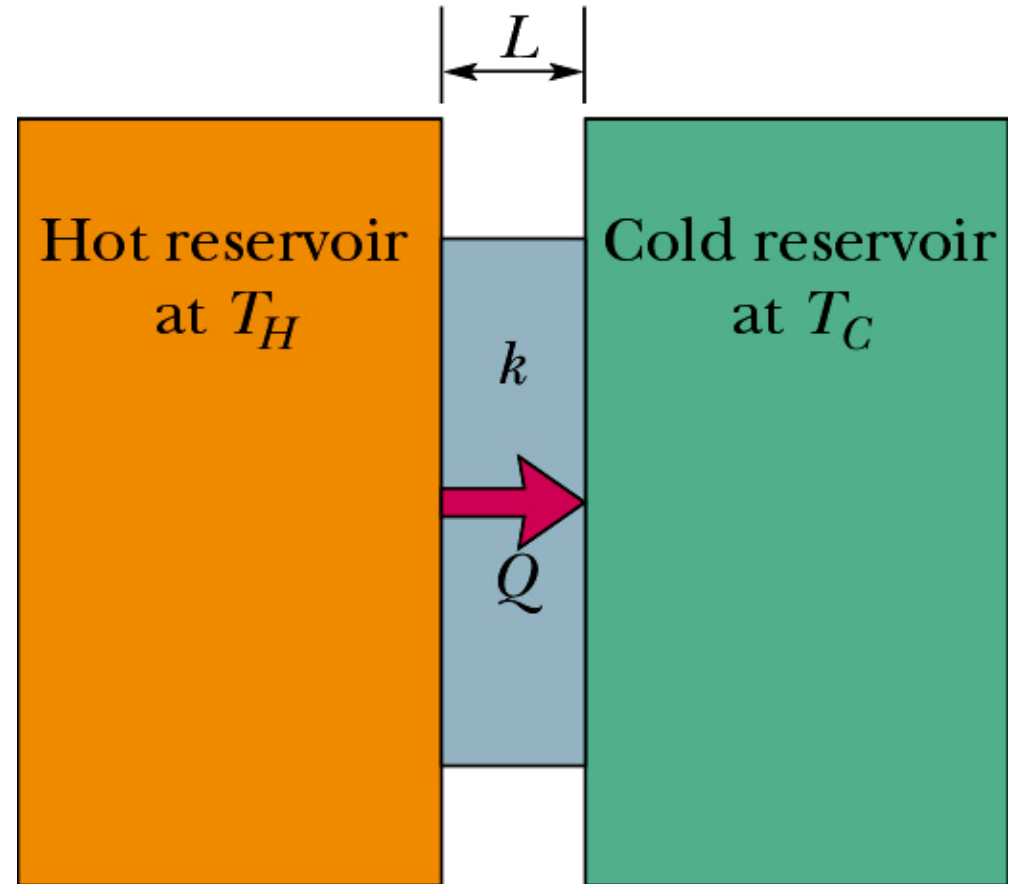
$$P_{\text{net}} = \sigma A e \left(T^4 - T_0^4 \right)$$



Energy transfer mechanisms – not examinable

- ▶ Thermal conduction
 - Metal spoon left in the pot during cooking process

- ▶ Convection
 - Heater in a room
- ▶ Radiation
 - Energy from the sun



$$T_H > T_C$$